

INTEGRAL TAIL SIGNATURES AND THE HILBERT–SMITH CONJECTURE

A. DABROWSKI'S AI AGENTS

ABSTRACT. We prove that the p -adic integers cannot act continuously and effectively on a connected finite-dimensional manifold. The obstruction is an integer signature in a category of asymptotically controlled rational equivariant modules. For finite cyclic p -groups acting through a fixed free C_p -space, a finite Mayer–Vietoris induction makes the difference between p copies of the identity and tensoring with the quotient permutation representation nilpotent. The signature of every four-dimensional class is therefore divisible by p at all sufficiently large indices. An effective p -adic action would yield a class of signature 1. Coordinate averaging gives an invariant map of degree one, and finite quotient representatives yield a homotopy idempotent compatible with Poincaré duality in a fixed exterior quotient. After forgetting equivariance and auxiliary control, an explicit homotopy isometry identifies its normalized summand with the local Poincaré complex of $S^n \times \mathbb{CP}^2$, where n is the manifold dimension. Successive localization boundaries give one positively oriented copy of $\mathbb{CP}^2$. The contradiction implies the Hilbert–Smith conjecture for second-countable locally compact groups.

1. INTRODUCTION

The Hilbert–Smith conjecture asserts that a locally compact group acting continuously and effectively on a connected finite-dimensional manifold is a Lie group. We prove the assertion for second-countable groups by excluding effective actions of the additive group of p -adic integers. The obstruction uses integer signatures of sequences of finite rational Poincaré complexes, considered up to eventual equality.

Theorem 1.1 (*p -adic exclusion*). *For every prime p , a jointly continuous action of $\mathbb{Z}_p$ on a connected Hausdorff, second-countable, finite-dimensional manifold, with or without boundary, has nontrivial kernel.*

Corollary 1.2 (Hilbert–Smith). *A second-countable locally compact Hausdorff group acting jointly continuously and effectively on such a manifold is a Lie group with its given topology.*

The manifold need not be compact, orientable, or triangulable, and continuity is the only regularity required of the action. The chain constructions use the compactification of the coordinate space of one chart, together with the auxiliary factor $\mathbb{CP}^2$.

The Hilbert–Smith conjecture is the transformation-group version of Hilbert’s fifth problem [Hil02]. The reduction to actions of the p -adic integers combines locally compact group structure with the rigidity of periodic homeomorphisms [New31, Smi41, Lee97]; a modern account, including a uniform form of Newman’s theorem, is [Pa19]. The conjecture is known in dimensions at most two [MZ55, Pa19] and in dimension three [Pa13]. In arbitrary dimension it holds for C^2 actions [BM46], Lipschitz actions [RS97], and $C^{0,\alpha}$ actions on n -manifolds with $\alpha > n/(n+2)$ [Mal97]. Martin announced the quasiconformal case [Mar99]. Mj extends this approach to boundaries of hyperbolic Poincaré duality groups under a Hausdorff dimension bound [Mj12]. Shelukhin studies a symplectic analogue [She24].

Yang established cohomological restrictions, including dimension raising, for p -adic orbit spaces [Yang60]. Our obstruction is an integer signature obtained from controlled algebra.

The proof has two parts. The algebraic argument uses a fixed compact free C_p -space T and finite cyclic groups G_i with index- p subgroups P_i . Let $\tau_i = \mathbb{Q}[G_i/P_i]$ with its positive permutation form. On the controlled L -groups over T , define

$$\nu = p \text{id} - \tau \otimes (-).$$

A covering trivialization locally identifies the tensor functor with p copies of the identity. Mayer–Vietoris for a finite closed cover makes ν nilpotent. Passing to the sequence of ordinary equivariant signature representations shows that every four-dimensional class has integer signatures divisible by p for all sufficiently large indices.

For the geometric construction, suppose that $\mathbb{Z}_p$ acts effectively. After restricting to a small open subgroup, coordinate averaging gives an invariant control map f to S^n , whose extension to the compactified coordinate space has degree one. A compact carrier avoiding the subgroup's fixed set maps equivariantly to one fixed free C_p -sphere T . Representatives of smaller finite quotients give chain maps on common fine triangulations. Their product and duality relations hold up to uniformly controlled homotopies and errors factoring through one fixed exterior. Adding free sheets cancels their graph displacement, and averaging produces a homotopy idempotent in the exterior quotient category. An explicit homotopy isometry identifies its normalized Poincaré summand with the scalar manifold germ.

The scalar comparison retains sphere control until successive localization boundaries have removed the exterior. Compatible relative fundamental cycles then compute the final class as one positively oriented copy of $\mathbb{CP}^2$, with signature 1. Taking the integer signature after these boundaries yields the contradiction $1 \in p\mathbb{Z}$.

Conventions. All group actions are jointly continuous. Manifolds are Hausdorff, second countable, and finite dimensional; Lie groups are finite dimensional and second countable. All chain coefficients are rational. Representation rings, Witt groups, L -groups, and the quotient $\prod_i \mathbb{Z} / \bigoplus_i \mathbb{Z}$ retain their integral abelian-group structure. We write L_* for ultimate lower quadratic L -theory unless a decoration is specified. Every bounded sequence of complexes has a single finite degree range, independent of the index. Equality of morphisms in an asymptotic category means eventual equality of their matrix representatives; equality in an exterior quotient means that the difference factors through an exterior-supported object by controlled maps. Small skeletons are implicit when a spectrum construction requires a small category.

2. ASYMPTOTIC CATEGORIES AND SUPPORT

Fix finite cyclic groups $G_i = C_{p^{a_i}}$ with $a_i \geq 1$, their index- p subgroups P_i , and identifications $G_i/P_i = C_p$. Let X be a compact metric C_p -space on which the action is isometric.

Definition 2.1. An object of $\mathcal{A}_{\mathbf{G}}(X)$ is a sequence

$$M_i = \mathbb{Q}[B_i], \quad \ell_i : B_i \longrightarrow X,$$

where B_i is a finite free G_i -set and ℓ_i is equivariant through $G_i \rightarrow C_p$. A morphism $u : M \rightarrow M'$ is an equivariant rational matrix sequence (u_i) , modulo eventual equality, satisfying

$$(2.1) \quad \text{prop}(u_i) = \sup_{(u_i)_{b' \neq 0}} d(\ell'_i(b'), \ell_i(b)) \longrightarrow 0.$$

Empty suprema are zero throughout. Duality fixes labeled bases and transposes matrices. The same construction with trivial groups defines the scalar category $\mathcal{A}_1(X)$.

Ranks and coefficient sizes are unrestricted. Composition and addition are well defined because

$$(2.2) \quad \text{prop}(vu) \leq \text{prop}(v) + \text{prop}(u), \quad \text{prop}(u + v) \leq \max\{\text{prop}(u), \text{prop}(v)\}.$$

Product control spaces have the maximum metric. Forgetting equivariance and pushing labels forward by a uniformly continuous equivariant map preserve duality.

For a closed invariant subset $S \subset X$, let $\mathcal{A}_{\mathbf{G},S}(X)$ be the full subcategory of objects with

$$\sup_{b \in B_i} d(\ell_i(b), S) \rightarrow 0.$$

For $S = \emptyset$ this is the zero category. When the group sequence is understood, we abbreviate this category to $\mathcal{A}_S$. Let I_S be the ideal of morphisms factoring through $\mathcal{A}_S$. For a fixed closed invariant exterior Z , put

$$(2.3) \quad \mathcal{B}_{\mathbf{G},Z}(X) = \mathcal{A}_{\mathbf{G}}(X) / \mathcal{A}_{\mathbf{G},Z}(X).$$

Thus an equation in this quotient means that its defect has a factorization through a Z -supported object by morphisms satisfying (2.1).

A useful matrix description of I_S is the following: $u \in I_S$ if and only if all source and target labels of its nonzero entries approach S uniformly. One implication follows from (2.2). For the other, restrict its source to the invariant based summand consisting of its nonzero columns. This gives a factorization through an S -supported free object.

Lemma 2.2 (Support filtration). *The inclusion $\mathcal{A}_S \subset \mathcal{A}_{\mathbf{G}}(X)$ is a duality-preserving Karoubi filtration in the sense of [CP95, Definition 1.27].*

Proof. For each nonnegative null sequence $r = (r_i)$, split an object M into the based summand $E_r M$ at distance at most r_i from S and its complement $U_r M$. The summands are equivariant, the first is S -supported, and their projections are transpose-invariant matrices of propagation zero. The maximum of two radii refines both splits.

If the labels of an S -supported object have distance at most a_i from S , and a morphism to or from M has propagation at most ε_i , every nonzero endpoint in M belongs to $E_{a+\varepsilon} M$. This is the required factorization. Taking maxima gives a common factorization for any finite family of maps. Finally, common-radius decompositions of $M \oplus M'$ are sums of the corresponding decompositions of M and M' , and dominate any pair of separate radii. These verify the four filtration axioms. Duality fixes the splits and interchanges the two factorization axioms. $\square$

Lemma 2.3 (Excision). *For a closed invariant cover $X = A \cup B$, with $Y = A \cap B$, one has*

$$(2.4) \quad \mathcal{A}_A \cap \mathcal{A}_B = \mathcal{A}_Y, \quad \mathcal{A}_A / \mathcal{A}_Y \simeq \mathcal{A}_{\mathbf{G}}(X) / \mathcal{A}_B.$$

The equivalence preserves involution.

Proof. For all sufficiently small $t > 0$, compactness gives $N_t(A) \cap N_t(B) \subset N_{\eta(t)}(Y)$ with $\eta(t) \rightarrow 0$. Otherwise points approaching both A and B but staying away from Y would have a limit in Y . This proves the first assertion.

Split each basis according to whether its label belongs to A or to $B \setminus A$. The second summand vanishes in the quotient by $\mathcal{A}_B$, proving essential surjectivity. If a morphism between A -supported objects factors through a B -supported object, retain only the intermediate basis rows reached by the first factor. Their distance from A is bounded by the source support distance plus the propagation of that factor. This intermediate summand is therefore both A - and B -supported, hence Y -supported. The two factorization ideals agree on $\mathcal{A}_A$, proving full faithfulness. If $Y = \emptyset$, compact separation makes the intermediate summand eventually zero. All basis splits are orthogonal, so the equivalence respects involution. $\square$

Lemma 2.4 (Separation). *If $d(S, Z) > 0$, the functor $\mathcal{A}_S \rightarrow \mathcal{B}_{\mathbf{G}, Z}(X)$ is fully faithful.*

Proof. A map between S -supported objects factoring through a Z -supported object has every nonzero endpoint approaching both separated sets. It is eventually zero. $\square$

Lemma 2.5 (Relabeling). *For nonempty closed invariant $S \subset X$, inclusion induces a duality-preserving equivalence $\mathcal{A}_{\mathbf{G}}(S) \simeq \mathcal{A}_{\mathbf{G}, S}(X)$.*

Proof. For each free basis orbit of an S -supported object, choose a nearest point of S to the representative label, and extend equivariantly. Labels already in S may be fixed. Isometry of the action makes the relabeled identity matrix asymptotically small. Leave matrix coefficients unchanged: the new propagation is bounded by the old propagation plus the two relabeling errors. Use identical choices for dual bases. If J denotes inclusion and F relabeling, the identity matrices define a duality-preserving natural isomorphism $JF \cong \text{id}$. $\square$

Support decompositions are performed on free labeled modules before idempotent completion. An object (M, e) of $\text{Kar}(\mathcal{A})$ carries the support of M .

3. INTEGER SIGNATURES AND THE TAIL CATEGORY

Let G be finite, V a finite-dimensional rational G -module, and b a nonsingular invariant symmetric form. Relative to an invariant positive inner product on $V \otimes \mathbb{R}$, the form b is represented by a self-adjoint operator. Its positive and negative spectral subspaces V^+ and V^- are G -invariant. Their complexifications define

$$(3.1) \quad \text{Sign}_G(V, b) = [V_{\mathbb{C}}^+] - [V_{\mathbb{C}}^-] \in R_{\mathbb{C}}(G).$$

This is independent of the chosen inner product and vanishes on metabolic forms. Indeed, let L be an invariant Lagrangian and choose an invariant complement K by averaging. The cross-pairing determines a unique equivariant map $c : K \rightarrow L$ with $b(cu, v) = b(u, v)$ for $u, v \in K$. Then $\{u - c(u)/2 : u \in K\}$ is an invariant Lagrangian complement to L . The resulting hyperbolic form has isomorphic positive and negative representations. Thus (3.1) defines the signature representation of a projective four-dimensional L -class. Its character at the identity is the ordinary integer signature. Tensoring with a positive-definite rational representation W multiplies this character by χ_W .

The passage from group-ring forms to invariant rational forms uses the *unnormalized* coefficient-of-identity trace. Explicitly,

$$(3.2) \quad \lambda(v, w) = \sum_{g \in G} b(v, gw)g, \quad \text{tr}_1 \lambda(v, w) = b(v, w).$$

For a free equivariant basis, group-ring adjoint is rational matrix transpose. Forgetting the action therefore preserves duality without a factor $|G|^{-1}$, also on projective summands.

Lemma 3.1 (Cyclic arithmetic). *Let $G = C_{p^a} = \langle g \rangle$ with $a \geq 1$. If $\text{Sign}_G(V, b)(g) = 0$, then $p \mid \sigma(V, b)$.*

Proof. Write $\text{Sign}_G(V, b) = \sum_{j=0}^{p^a-1} m_j \chi_j$, with $m_j \in \mathbb{Z}$ and $\chi_j(g) = \zeta_{p^a}^j$. Then $F(t) = \sum_j m_j t^j$ vanishes at ζ_{p^a} , so the monic minimal polynomial Φ_{p^a} divides F in $\mathbb{Z}[t]$. Consequently $\sigma(V, b) = F(1)$ is divisible by $\Phi_{p^a}(1) = p$. $\square$

Forgetting labels defines a duality-preserving functor to the tail category

$$(3.3) \quad \mathcal{V}_{\mathbf{G}} = \left(\prod_i \text{Free}(\mathbb{Q}[G_i]) \right) / \left(\bigoplus_i \text{Free}(\mathbb{Q}[G_i]) \right),$$

where the direct sum denotes the full subcategory of objects supported on finitely many indices. Its morphisms are sequences of $\mathbb{Q}[G_i]$ -module maps modulo eventual equality.

Proposition 3.2 (Tail extraction). *There is a natural isomorphism*

$$(3.4) \quad L_4(\mathcal{V}_{\mathbf{G}}) \cong \frac{\prod_i L_4^p(\mathbb{Q}[G_i])}{\bigoplus_i L_4^p(\mathbb{Q}[G_i])}.$$

It commutes with forgetting equivariance and with tensoring by sequences of finite positive equivariant forms.

Proof. The rings $\mathbb{Q}[G_i]$ are semisimple artinian, hence regular noetherian of global dimension zero. Thus $K_{-k}(\mathbb{Q}[G_i]) = 0$ for $k \geq 1$ [Wei13, Chapter III, Section 4]. The same statement for the free factor categories follows from [Ran92, Section 11, p. 101]. The cutoff is independent of i . The countable product theorem [CP95, proof of Theorem 5.4], [Ros06, Section 2.1(2)] therefore gives

$$\mathbf{L}^{\langle -\infty \rangle} \left(\prod_i \text{Free}(\mathbb{Q}[G_i]) \right) \simeq \prod_i \mathbf{L}^{\langle -\infty \rangle}(\text{Free}(\mathbb{Q}[G_i])).$$

Let $\mathcal{S}$ be the finite-support subcategory in (3.3). Splitting at finite index sets makes it a Karoubi filtration. Its negative K -groups are

$$K_{-k}(\mathcal{S}) = \bigoplus_i K_{-k}(\text{Free}(\mathbb{Q}[G_i])) = 0 \quad (k \geq 1).$$

The finite-support assertion holds in the Bass polynomial models: an idempotent, a polynomial matrix, and any finite list of defining relations lie in one finite block of indices. Projection onto that block also carries any relation in a larger block back to the original block. Thus the resulting cokernels are exactly the direct sums of the factor cokernels.

The Rothenberg sequences [Ran92, Theorem 17.2] identify projective and ultimate decoration for $\mathcal{S}$ and for each semisimple factor. Finite projective Poincaré data in $\text{Kar}(\mathcal{S})$ have finite index support. In degrees 4 and 3, the map from $\mathcal{S}$ to the product is therefore the injective map

$$\bigoplus_i L_j^p(\mathbb{Q}[G_i]) \longrightarrow \prod_i L_j^p(\mathbb{Q}[G_i]).$$

Localization [CP95, Theorem 4.2] identifies the degree-four group of the quotient with its cokernel, proving (3.4). Every comparison is natural for the stated duality-preserving functors. $\square$

By algebraic surgery below the middle dimension [Ran89, Theorem 5.2] and the rational half-unit comparison, an element of $L_4^p(\mathbb{Q}[G_i])$ is represented by a finite nonsingular rational equivariant form in degree two. Alternatively, semisimplicity splits a Poincaré complex onto homology, leaving hyperbolic pairs off the middle degree. Hence proposition 3.2 assigns to a controlled class a tail of finite forms. Equality of tails means equality of the factor L -classes for every sufficiently large i ; taking signatures gives a well-defined homomorphism

$$(3.5) \quad \sigma_{\text{tail}} : L_4(\mathcal{A}_{\mathbf{G}}(X)) \longrightarrow \mathbb{Z}^{\mathbb{N}} / \mathbb{Z}^{(\mathbb{N})}.$$

In the geometric argument, this homomorphism is applied after localization boundaries have removed the exterior.

4. SIGNATURE DIVISIBILITY OVER A FREE C_p -SPACE

Let T be a fixed nonempty compact free C_p -space, equipped with a compatible C_p -invariant metric. Equip $\tau_i = \mathbb{Q}[G_i/P_i]$ with its orthonormal positive form. Tensoring with τ_i , with diagonal action and the original spatial label, preserves the category and its duality. Define

$$(4.1) \quad \nu = p \text{ id} - \tau \otimes (-) \quad \text{on } L_*(\mathcal{A}_{\mathbf{G}}(T)).$$

The permutation $(a, b) \mapsto (a, a^{-1}b)$ gives an orthogonal equivariant isomorphism $\tau_i \otimes \tau_i \cong \tau_i^{\oplus p}$; in particular $\nu^s = p^{s-1}\nu$ for $s \geq 1$, as an equality of integral endomorphisms.

Lemma 4.1 (Local triviality). *If $U \subset T/C_p$ is compactly contained in an open covering trivialization and T_U is its inverse image, then $\nu = 0$ on the L -groups $L_*(\mathcal{A}_{\mathbf{G}, T_U}(T))$.*

Proof. Choose a closed invariant neighborhood L of T_U inside the open lift of the trivialization. Let c be its locally constant sheet function, so $c(hx) = \bar{h}c(x)$. Labels of a T_U -supported object eventually belong to L . The map

$$(4.2) \quad e_b \otimes m_x \mapsto m_x \quad \text{in copy } c(x)^{-1}b \in C_p$$

is an equivariant orthogonal isomorphism of propagation zero. The finitely many sheet pieces of L are disjoint compact subsets, hence positively separated. A morphism of propagation tending to zero eventually has no entry between distinct pieces. Thus (4.2) is natural in the tail category. It extends to idempotent completion and to all lower decorations, giving $\tau \otimes (-) = p \text{ id}$ on these L -groups. $\square$

Choose a finite closed cover $U_1, \dots, U_s$ of T/C_p , with each U_a compactly contained in an open trivialization, and let T_a be its inverse image. This cover and s are fixed before the groups vary.

Lemma 4.2 (Nilpotence). *On $L_*(\mathcal{A}_{\mathbf{G}}(T))$ one has $\nu^s = 0$.*

Proof. Induct on s . The case $s = 1$ is lemma 4.1. Put $A = T_1 \cup \dots \cup T_{s-1}$ and $B = T_s$. The support Mayer–Vietoris sequence obtained from lemmas 2.2 and 2.3 and localization is natural for $\tau \otimes (-)$. On $A \cap B$, $\nu = 0$ by lemma 4.1. Therefore the boundary of νx is zero, and exactness writes νx as an image of a sum of classes from the A - and B -support categories. Relabeling lemma 2.5 identifies the first with the category on A ; the induction hypothesis annihilates it by ν^{s-1} . Local triviality annihilates the second by ν , and therefore also by ν^{s-1} when $s > 1$. Applying ν^{s-1} proves the claim. $\square$

Theorem 4.3 (Integer tail divisibility). *For every $\alpha \in L_4(\mathcal{A}_{\mathbf{G}}(T))$, its integer signature tail is eventually divisible by p .*

Proof. Use proposition 3.2 to represent the extracted tail by forms (V_i, b_i) . Naturality and lemma 4.2 give

$$(4.3) \quad (p - \tau_i)^s [V_i, b_i] = 0 \quad \text{in } L_4^p(\mathbb{Q}[G_i]) \quad \text{for all sufficiently large } i.$$

For $g \notin P_i$, the permutation character of τ_i is zero. Taking signature characters in (4.3) gives $p^s \text{Sign}_{G_i}(V_i, b_i)(g) = 0$. Hence this character value is zero. A generator of G_i is outside P_i , so lemma 3.1 gives $p \mid \sigma(V_i, b_i)$ for all sufficiently large i . $\square$

5. RATIONAL POINCARÉ CORNERS

We construct the Poincaré summand of a compatible homotopy idempotent and determine the normalization of its scalar form. The construction will be applied in an exterior quotient.

5.1. Signs and the rational structure complexes. Complexes are homologically graded. Our N -shifted duality is

$$(5.1) \quad (C^{N-*})_r = C_{N-r}^*, \quad d_{C^{N-*},r} = (-1)^r d_C^*, \quad (T\phi)_r = (-1)^{r(N-r)} \phi_{N-r}^*.$$

The double shifted dual has differential $(-1)^{N+1} d_C$; its canonical identification with C is multiplication in degree r by $\epsilon_r = (-1)^{r(N-r)} = (-1)^{r(N+1)}$. This identification is part of the convention. Thus $T^2 = 1$, and conjugation by a degree-zero chain map commutes with T .

For $t = \sum x_r \otimes y_{N-r}$, right slant is

$$\lambda_N(t)_r(\alpha) = \sum x_r \alpha(y_{N-r}),$$

with no additional degree sign. The tensor differential is $d(x_r \otimes y) = dx_r \otimes y + (-1)^r x_r \otimes dy$; graded tensor transposition induces (5.1).

For a bounded complex over a rational additive category, use the signed Hom complex

$$M(C)_k = \bigoplus_r \text{Hom}(C_{k-r}^*, C_r), \quad (\partial f)_r = d_C f_{r+1} + (-1)^r f_r d_C^*.$$

For $f \in M(C)_k$, transposition is $(Tf)_r = (-1)^{r(k-r)} f_{k-r}^*$. For finite scalar based rational complexes, right slant identifies $M(C)$ with $C \otimes C$. In equivariant and controlled categories, the same definition uses the morphism groups of that category. Since C_2 -invariants and coinvariants are exact over $\mathbb{Q}$, the symmetric and quadratic structure groups are

$$(5.2) \quad Q^N(C) = H_N(M(C)^{C_2}), \quad Q_N(C) = H_N(M(C)_{C_2}).$$

The norm is $1 + T$; its inverse sends an invariant cycle t to the coinvariant cycle $t/2$. Averaging a primitive proves the same statement for boundaries. In particular a symmetric structure may be represented by a strict symmetric chain map ϕ , with higher components zero; its quadratic lift has $\psi_0 = \phi/2$ and $\psi_s = 0$ for $s > 0$. The same argument applies to relative structure complexes, whose cone maps commute with transposition [Ran80I, Proposition 3.3], [Ran89, Section 3]. These operations preserve propagation and exterior factorizations.

5.2. Splitting a compatible idempotent. Suppose that a finite cyclic group G , of order q , acts on C up to chain homotopy by maps satisfying

$$(5.3) \quad A_g A_h \simeq A_{gh}, \quad A_1 = 1, \quad A_g \phi \simeq \phi A_{g^{-1}}^*.$$

On $\mathcal{C} = C \otimes \mathbb{Q}[G]$ put

$$(5.4) \quad \Phi = \phi \otimes 1, \quad E = q^{-1} \sum_{g \in G} A_g \otimes R_{g^{-1}},$$

where right translation sends sheet b to bg^{-1} and the group acts strictly by left translation. Multiplication gives

$$(5.5) \quad E^2 \simeq E, \quad E\Phi \simeq \Phi E^*, \quad E_{ab} = q^{-1} A_{a^{-1}b}.$$

In the controlled application, the individual A_g become small after forgetting the auxiliary coordinate. The compensated matrices E and Φ , together with the homotopies in (5.5), will be small before that projection; this is proved in §11. The following lemma applies in that quotient category.

Lemma 5.1 (Compatible splitting). *Let $\mathcal{Q}$ be a rational additive category with involution and let $\Phi : \mathcal{C}^{N-*} \rightarrow \mathcal{C}$ be a strict symmetric Poincaré equivalence. Suppose $\mathcal{C}$ is concentrated in*

$[0, N]$, q is an invertible self-adjoint central coefficient, and E is a chain map with specified homotopies $E^2 \simeq E$ and $E\Phi \simeq \Phi E^*$ in $\mathcal{Q}$. Then E splits in $K^b(\text{Kar } \mathcal{Q})$ as

$$(5.6) \quad D \xrightarrow{\iota} \mathcal{C} \xrightarrow{r} D, \quad \iota r \simeq E, \quad r \iota \simeq 1,$$

with D concentrated in $[0, N]$. Its normalized Poincaré duality and a homotopy inverse are

$$(5.7) \quad \phi_D = q^{-1} r \Phi r^*, \quad b_D = q \iota^* \Phi^{-1} \iota,$$

where Φ^{-1} denotes a chosen chain homotopy inverse. This construction determines a class in $L_N(\mathcal{Q})$ naturally under duality-preserving functors.

Proof. Equip $\text{Kar } \mathcal{Q}$ with its split exact structure. Its bounded acyclic complexes are contractible, so its bounded derived category is $K^b(\text{Kar } \mathcal{Q})$. This category is idempotent complete by [BS01, Theorem 2.8], giving (5.6).

If D has degrees below zero, the identity homotopy $r \iota \simeq 1$ gives a split surjection from the next degree onto its lowest term; above N it gives a split injection from the highest term to its predecessor. Remove the resulting contractible two-term summands. Their complements belong to $\text{Kar } \mathcal{Q}$. Finitely many removals leave D in $[0, N]$.

The compatibility homotopy implies

$$\phi_D b_D \simeq r \Phi E^* \Phi^{-1} \iota \simeq r E \iota \simeq 1, \quad b_D \phi_D \simeq \iota^* \Phi^{-1} E \Phi r^* \simeq \iota^* E^* r^* \simeq 1.$$

Conjugation by r respects signed transposition, so ϕ_D is strict symmetric. By section 5.1, its half represents the full quadratic structure.

For two splittings, $r_2 \iota_1$ and $r_1 \iota_2$ are inverse chain equivalences and conjugate the symmetric structures up to homotopy. Averaging the primitive of that difference, then taking its half, identifies their full quadratic structure classes. The Poincaré mapping-cylinder cobordism [Ran89, Definition 3.10 and the following paragraph] identifies their L -classes.

Finally, the inclusion $j : \mathcal{Q} \rightarrow \text{Kar } \mathcal{Q}$ induces

$$j_* : L_N(\mathcal{Q}) \xrightarrow{\cong} L_N(\text{Kar } \mathcal{Q})$$

by ultimate completion invariance [Ran92, Definition 17.7]. Map the constructed class to ultimate decoration and apply j_*^{-1} . The resulting class is natural because j_* is natural. $\square$

When $\mathcal{Q}$ is an asymptotic exterior category, the splitting is performed in $K^b(\text{Kar } \mathcal{Q})$. Its maps and homotopies are therefore represented by sequences of propagation tending to zero, and their equations have controlled factorizations through exterior-supported objects. The number of degrees, maps, and equations is finite.

Lemma 5.2 (Scalar normalization). *For the corner of (5.4), suppose a duality-preserving forgetting functor retains a coordinate in which all A_g and the homotopies in (5.3) are uniformly small. After applying that functor, the normalized corner (5.7) is homotopy-isometric to (C, ϕ) .*

Proof. Let j_1 include the identity sheet and π_1 project to it. Set

$$v = E j_1, \quad u = q \pi_1 E, \quad v_a = q^{-1} A_{a-1}, \quad u_b = A_b.$$

These row and column maps are used only after the indicated forgetting functor. From (5.3),

$$(5.8) \quad uv \simeq 1, \quad vu \simeq E, \quad uE \simeq u, \quad Ev \simeq v, \quad u\Phi u^* = \sum_b A_b \phi A_b^* \simeq q\phi.$$

For $w = u\iota : D \rightarrow C$ and $z = r\nu : C \rightarrow D$, these identities give $wz \simeq 1$, $zw \simeq 1$, and

$$(5.9) \quad w\phi_D w^* \simeq q^{-1} u E \Phi E^* u^* \simeq q^{-1} u \Phi u^* \simeq \phi.$$

Average a primitive for the difference in (5.9), then divide it by two. This identifies the complete symmetric and quadratic structure classes. The Poincaré mapping-cylinder cobordism identifies the ordinary L -classes in the completed category, and ultimate completion invariance identifies them in the original category.

For $q = q_i$ varying with the index, multiplication by q_i or q_i^{-1} has propagation zero. Uniform smallness of the entries and homotopies makes the row and column maps valid sequence morphisms. Thus the normalization is realized by the homotopy isometry w in (5.9). $\square$

6. LOCALIZATION BOUNDARIES

Let $X = A \cup B$ be a closed invariant cover, and put $Y = A \cap B$. The support filtration and excision equivalence established above, together with [CP95, Theorem 4.2], give the Mayer–Vietoris connecting homomorphism

$$(6.1) \quad \partial_{A,B} : L_k(\mathcal{A}(X)) \longrightarrow L_{k-1}(\mathcal{A}_Y(X)).$$

Here and below the group sequence is suppressed when it is unchanged. The exterior version takes values in the same support category.

Proposition 6.1. *If $Z \subset A$ is closed and invariant and $d(Z, B) > 0$, there is a natural Mayer–Vietoris boundary*

$$(6.2) \quad \partial_{A,B} : L_k(\mathcal{A}(X)/\mathcal{A}_Z(X)) \longrightarrow L_{k-1}(\mathcal{A}_Y(X)).$$

It commutes with duality-preserving functors preserving the support subcategories.

Proof. Set $\mathcal{Q} = \mathcal{A}(X)/\mathcal{A}_Z(X)$ and $\mathcal{U} = \mathcal{A}_A(X)/\mathcal{A}_Z(X)$. Separation embeds the B - and Y -support categories fully faithfully in $\mathcal{Q}$. Choose controlled representatives of a finite family of quotient maps, and make the based support decompositions before quotienting. The support filtration argument then applies in the quotient. The based split into the A - and $B \setminus A$ -parts gives the duality-preserving excision equivalence

$$(6.3) \quad \mathcal{U}/\mathcal{A}_Y(X) \simeq \mathcal{Q}/\mathcal{A}_B(X).$$

Indeed, the right side kills $I_Z + I_B$. On A -supported objects the restriction of I_B is I_Y , by the support excision proof, while I_Z already factors through A -supported objects. The left side therefore kills exactly $I_Z + I_Y$. The inclusion and projection of the based split are mutual adjoints and are inverse after the indicated quotient.

The boundary is the composite induced on homotopy groups by

$$\mathbf{L}^{\langle -\infty \rangle}(\mathcal{Q}) \longrightarrow \mathbf{L}^{\langle -\infty \rangle}(\mathcal{Q}/\mathcal{A}_B) \simeq \mathbf{L}^{\langle -\infty \rangle}(\mathcal{U}/\mathcal{A}_Y) \longrightarrow \Sigma \mathbf{L}^{\langle -\infty \rangle}(\mathcal{A}_Y).$$

Localization for the two filtrations gives the Mayer–Vietoris sequence. Naturality follows from the same localization diagram. $\square$

The next lemma computes the boundary of a union of Poincaré pairs. The common boundary is oriented as the boundary of the A -side.

Lemma 6.2. *Let $V_A = (P \rightarrow D_A)$ and $V_B = (P \rightarrow D_B)$ be k -dimensional quadratic Poincaré pairs with oppositely oriented common boundary P . Suppose D_A, D_B, P are supported on A, B, Y , respectively, and put $V = V_A \cup_P V_B$. Then*

$$(6.4) \quad \partial_{A,B}[V] = [P].$$

The assertion holds for (6.1), and also for (6.2) when the union represents a class after exterior quotienting.

Proof. In the ordinary case put $\mathcal{Q} = \mathcal{A}(X)$ and $\mathcal{U} = \mathcal{A}_A(X)$; in the exterior case use the categories defined in the preceding proof. The A -pair $P \rightarrow D_A$ is a null-cobordism of P in $\mathcal{U}$. In the relative Poincaré-pair model it represents a class of the cofiber of $\mathbf{L}^{(-\infty)}(\mathcal{A}_Y) \rightarrow \mathbf{L}^{(-\infty)}(\mathcal{U})$, with connecting image $[P]$. Its image in $\mathcal{U}/\mathcal{A}_Y$ is the closed complex D_A with the full induced quadratic structure, since the boundary complex has become zero.

In the quotient of $\mathcal{Q}$ by the B -support category, the B -top, P , and the suspended interface summand in the union vanish degree-wise. All interface corrections in the union structure vanish with them. The surviving differential and every structure component are precisely those on D_A . Excision identifies these representatives. The relative-to-quotient comparison is the one in [CP95, proof of Theorem 4.1]; passing to ultimate decoration gives the diagram above. Its connecting image is consequently $[P]$. Additive completion and lower decoration preserve the pair and this diagram. $\square$

7. LOCAL CHAIN CARRIERS

The carrier lemmas below provide the control estimates for the closed and relative Poincaré constructions. These estimates depend on support rather than coefficient norms.

Lemma 7.1 (Finite acyclic carriers). *Let E be a finite based chain complex concentrated in nonnegative degrees, and let F be a chain complex over $\mathbb{Q}$. For every basis element $e \in E_r$, suppose that a chain subcomplex $\Gamma(e) \subset F$ has been specified. Assume*

- (1) $\Gamma(e') \subset \Gamma(e)$ whenever the coefficient of e' in de is nonzero;
- (2) each $\Gamma(e)$ has an augmentation to $\mathbb{Q}$ making its augmented chain complex acyclic, and these augmentations agree on inclusions of carriers.

If chain maps $u, v : E \rightarrow F$ are carried by Γ , and their images of each degree-zero basis element have the same augmentation, then there is a chain homotopy H with

$$u - v = dH + Hd, \quad H(e) \in \Gamma(e)_{r+1}.$$

Proof. Set $H_{-1} = 0$. Suppose H has been defined below degree r . For a basis element $e \in E_r$, put

$$w_e = u(e) - v(e) - H(de).$$

The nesting hypothesis puts this chain in $\Gamma(e)$. The chain-map identities and the equations already established in lower degrees give $dw_e = 0$. In degree zero its augmentation is zero. Acyclicity of the augmented carrier therefore gives $H(e)$ in that carrier with $dH(e) = w_e$. Extend linearly, and continue through the finitely many degrees of E . $\square$

The same induction constructs a carried chain map from an augmented source complex when an augmentation-compatible image for each degree-zero basis element is prescribed. These assertions hold equally for chain complexes with signed cellular bases: the augmentation is the geometric augmentation, not necessarily the sum of the coordinates in the selected basis.

Corollary 7.2 (Metric form of the carrier lemma). *Suppose the complexes in Lemma 7.1 have basis labels in a metric space X . If every basis label occurring in a chain of $\Gamma(e)$ is at distance at most a from the source label of e , the resulting homotopy has propagation at most a . If the labels are obtained by applying a fixed continuous map c on a compact metric space and every carrier lies within distance a of the geometric location of e , its propagation is at most $\omega_c(a)$, for any increasing continuity modulus ω_c of c .*

Proof. The recursive choices introduce no cells outside the assigned carrier. Apply the stated label bound to every nonzero matrix coefficient. $\square$

Lemma 7.3 (Filling on a fixed triangulation). *Let K triangulate a compact manifold W , let its mesh in a compatible metric be at most η , and let $C \subset W$ be contained in one coordinate chart and convex in that chart. Let $\gamma \in C_r(K; \mathbb{Q})$ be a cycle such that every closed simplex occurring in γ is contained in C . For $r = 0$, assume additionally that γ has augmentation zero. Then there is $q \in C_{r+1}(K; \mathbb{Q})$ such that*

$$dq = \gamma, \quad \bigcup_{\sigma \text{ occurring in } q} |\sigma| \subset N_\eta(C).$$

In particular, when $r = \dim W$, these hypotheses force $\gamma = 0$. Subdivision is confined to an auxiliary cone, so the target triangulation and its prescribed chain maps remain fixed.

Proof. Let $A \subset K$ be the subcomplex generated by the simplices occurring in γ . Its underlying space is contained in C . Contract its geometric inclusion to a point of C by a straight-line homotopy in the selected chart. This defines a continuous map

$$F : |\text{cone } A| \longrightarrow W$$

whose restriction to A is its original inclusion and whose entire image lies in C . After sufficiently many barycentric subdivisions of this auxiliary cone, the simplicial approximation theorem supplies a simplicial map $t : L \rightarrow K$. Let $S : C_*(\text{cone } A) \rightarrow C_*(L)$ be subdivision.

The restriction of tS to $C_*(A)$ is exactly the identity. To see this, if x is a vertex of the subdivided base, the approximation condition gives $x = F(x) \in \text{st}_K^\circ(t(x))$. Hence $t(x)$ is a vertex of the smallest original simplex containing x . Thus tS is carried in each original base simplex and fixes original vertices. Induction on dimension makes it the identity: after the equality on the boundary of an r -simplex, the difference is an r -cycle in that simplex; for $r > 0$ the boundary map on its one-dimensional group of top chains is injective.

In the augmented cone complex, choose the usual cone operator c ; for the given reduced cycle, $d(c\gamma) = \gamma$. Set $q = tS(c\gamma)$. The preceding base identity proves $dq = \gamma$. To estimate its support, take a point z in the relative interior of a source simplex of L . The approximation condition puts $F(z)$ in the open star of the image of each vertex of that source simplex. Consequently its image simplex is a face of the smallest target simplex containing $F(z)$. That target simplex meets C and has diameter at most η . The entire target simplex, and hence its image face, lies in $N_\eta(C)$. This proves the claim. If $r = \dim W$, then $q = 0$ and the already proved equality forces $\gamma = 0$. $\square$

Lemma 7.4 (Relative dual-block carrier maps). *Let A be a compact oriented combinatorial m -manifold with boundary, and let K be a finite triangulation with ∂A a subcomplex and mesh η . For $\tau \not\subset \partial A$, orient its dual block by*

$$\partial D\tau = (-1)^{\dim \tau + 1} \sum_{\substack{v \supset \tau \\ \dim v = \dim \tau + 1}} [v : \tau] Dv.$$

For a top simplex ρ with oriented fundamental coefficient ϵ_ρ , set $D\rho = \epsilon_\rho[b_\rho]$. Let D_{rel} be the cellular chain complex generated by the dual blocks $D\tau$ for simplices τ not contained in ∂A . It is a subcomplex of the barycentric dual cellular complex of A , including the dual cells on ∂A . For each simplex σ put

$$J_\sigma = \bigcup_{\substack{\tau \supset \sigma \\ \tau \not\subset \partial A}} D\tau.$$

The geometric augmentation on every J_σ makes its augmented cellular chain complex acyclic. There are chain maps

$$S_{\text{rel}} : D_{\text{rel}} \longrightarrow C_*(\text{sd } K), \quad R_{\text{rel}} : C_*(\text{sd } K) \longrightarrow D_{\text{rel}}$$

such that $R_{\text{rel}}S_{\text{rel}} = 1$ exactly and $S_{\text{rel}}R_{\text{rel}} \simeq 1$. Their matrices and the latter homotopy have geometric propagation at most $C_m\eta$, where C_m is independent of the number of simplices and all matrix dimensions.

Proof. The cellular boundary of $D\tau$ involves only blocks of proper cofaces of τ . Every such coface is still not contained in the boundary, proving the subcomplex assertion. The carriers satisfy $J_\tau \subset J_\sigma$ whenever $\tau \supseteq \sigma$.

If σ is not contained in ∂A , its dual block is a ball and $J_\sigma = D\sigma$. If $\sigma \subset \partial A$, write $l = m - \dim \sigma - 1$. Its link is a combinatorial l -ball. Nonboundary cofaces of σ correspond to nonboundary faces of this link. If $q = \dim \sigma$ and cofaces have join orientations, then $[\sigma * \lambda' : \sigma * \lambda] = (-1)^{q+1}[\lambda' : \lambda]$. The dual-block boundary formula therefore gives a chain isomorphism

$$C^{l-*}(\text{link}_A \sigma, \partial \text{link}_A \sigma; \mathbb{Q}) \longrightarrow C_*(J_\sigma; \mathbb{Q}), \quad \lambda^* \longmapsto (-1)^{r(l+1)} D(\sigma * \lambda)$$

in degree r , with the shifted-dual differential of (5.1). Its homology is thus $\mathbb{Q}$ in degree zero and zero in positive degrees. The geometric augmentation is nonzero on every top-coface dual vertex, so it identifies this degree-zero homology with $\mathbb{Q}$. This proves augmented acyclicity, including the zero-dimensional link case.

For a barycentric simplex with flag $\sigma_0 \subsetneq \cdots \subsetneq \sigma_r$, use J_{σ_0} as its carrier. On deleting the minimum face this carrier shrinks; on deleting any other face it remains unchanged. Choose an augmentation-one zero chain in each J_σ . The existence part of Lemma 7.1 constructs R_{rel} in these carriers. Let S_{rel} be cellular subdivision of dual blocks. For each interior dual block $D\sigma$, the composite $R_{\text{rel}}S_{\text{rel}}$ is carried inside $D\sigma$. Augmentation makes this composite the identity on dual vertices. Induction on the dimensions of the dual cells proves $R_{\text{rel}}S_{\text{rel}} = 1$: the residual on a cell is a top-dimensional cycle in its dual ball, hence zero.

For the other composite, both $S_{\text{rel}}R_{\text{rel}}$ and the identity on a barycentric flag with minimum σ are carried in $\text{sd}(\text{st}_K \sigma)$. These are nested augmented acyclic carriers because closed stars in a combinatorial manifold are balls. Lemma 7.1 supplies its homotopy. Every point of any such closed star is at distance at most η from any fixed point of σ , and the diameter of the closed star is at most 2η . The same estimate applies to J_σ and its subdivided cells. Corollary 7.2 gives the claimed uniform metric estimate. $\square$

Lemma 7.5 (Controlled cap-block comparison). *In the setting of Lemma 7.4, put $R = C_*(A, \partial A; \mathbb{Q})$ and use the homological shifted dual R^{m-*} with differential $(-1)^r d^*$ in degree r . Let $\mathcal{J} : R^{m-*} \rightarrow D_{\text{rel}}$, defined by $\mathcal{J}_r(\sigma^*) = (-1)^{r(m+1)} D\sigma$, be the signed dual-block identification. Let $S_K : C_*(K) \rightarrow C_*(\text{sd } K)$ and R_K be the usual simplicial subdivision map and a simplex-carried inverse, with $R_K S_K = 1$. Define*

$$F_{\text{block}} = R_K S_{\text{rel}} \mathcal{J}, \quad B_{\text{block}} = \mathcal{J}^{-1} R_{\text{rel}} S_K.$$

Suppose $F_{\text{cap}} : R^{m-*} \rightarrow C_*(K)$ is an exact chain map such that

- (1) for every nonboundary simplex σ , the image of σ^* is carried in the closed coface star $\text{st}_K \sigma$;
- (2) for each top simplex ρ , the degree-zero chain $F_{\text{cap}}(\rho^*)$ has geometric augmentation equal to the oriented coefficient of ρ in the fundamental chain of A .

Then F_{cap} is homotopic to F_{block} by maps of propagation at most $C_m\eta$. It has the homotopy inverse B_{block} , with both inverse homotopies of propagation at most $C'_m\eta$. After labeling by a fixed continuous control map c on A , all these propagations tend to zero as $\eta \rightarrow 0$.

Proof. The signed dual-block identification is a chain isomorphism. The subdivision identities and the two controlled subdivision homotopies show that F_{block} and B_{block} are inverse up to

homotopy. Explicitly, in their two composites replace respectively $S_{\text{rel}}R_{\text{rel}}$ and $S_K R_K$ by the identity; the remaining composites are $R_K S_K = 1$ and $\mathcal{J}^{-1}R_{\text{rel}}S_{\text{rel}}\mathcal{J} = 1$. Each replacement is a fixed finite composition of maps of geometric propagation $O_m(\eta)$.

The differential of R^{m-*} on a basis element σ^* uses only proper cofaces τ^* , and $\text{st}_K \tau \subset \text{st}_K \sigma$. Consequently the closed coface stars are nested augmented acyclic carriers for this source complex. The two displayed hypotheses on F_{cap} also hold for F_{block} , with the same augmentations. Apply Lemma 7.1 to obtain their exact homotopy in these stars. Composing this homotopy with B_{block} adjusts both already constructed inverse homotopies. Only a fixed finite number of compositions is added. Closed stars have diameter at most 2η , so the stated constants are independent of rank and valence. Apply the continuity modulus of c to conclude the last assertion. $\square$

Corollary 7.6 (Application to a common simplicial diagonal). *Let $j : C_*(\partial A; \mathbb{Q}) \rightarrow C_*(A; \mathbb{Q})$ and $q : C_*(A; \mathbb{Q}) \rightarrow C_*(A, \partial A; \mathbb{Q})$ denote inclusion and quotient. Choose one vertex order on K and use its Alexander–Whitney diagonal on A and all specified subcomplexes. If z_A is an oriented fundamental chain with $dz_A = jz_{\partial A}$, put*

$$t_A = \frac{1}{2}(1 + T)\Delta z_A, \quad t_{\partial A} = \frac{1}{2}(1 + T)\Delta z_{\partial A}.$$

They satisfy the boundary equation $dt_A = j^{\otimes 2}t_{\partial A}$. The right slant of t_A , restricted to relative cochains, satisfies the hypotheses of Lemma 7.5; it therefore has a controlled chain inverse and both controlled inverse homotopies. If ∂A is empty, this proves the corresponding closed cap-duality statement.

Proof. The diagonal is a chain map and restricts to every subcomplex, so the boundary equation holds before passage to any quotient. Write ϕ_A and $\phi_{\partial A}$ for the right slants of t_A and $t_{\partial A}$. They satisfy

$$d\phi_{A,r} - (-1)^r \phi_{A,r-1}d^* = j\phi_{\partial A,r-1}j^*.$$

The right side vanishes on relative cochains because $j^*q^* = 0$. Thus the relative slant is an exact chain map. Each term of the ordered diagonal of a simplex is a tensor of two faces of that simplex; its graded flip has the same property. On slanting against σ^* , every output face is therefore in a coface of σ . This gives the required coface-star support. Finally, for a top simplex $\rho = [v_0, \dots, v_m]$ with fundamental coefficient ϵ_ρ , the degree-zero image of ρ^* is

$$\frac{1}{2}\epsilon_\rho([v_0] + [v_m]),$$

with geometric augmentation ϵ_ρ . The required hypotheses are now verified, and the preceding lemma applies. $\square$

Lemma 7.7 (Exact comparison of diagonals on unchanged models). *Let $S : C_*(K) \rightarrow C_*(L)$ be subdivision and let $t : L \rightarrow K'$ be a simplicial map. Equip each triangulation with an ordered simplicial diagonal. There are chain homotopies satisfying*

$$\begin{aligned} dJ_S + J_S d &= (S \otimes S)\Delta_K - \Delta_L S, \\ dJ_t + J_t d &= (t \otimes t)\Delta_L - \Delta_{K'} t. \end{aligned}$$

The first is carried, on a simplex σ of K , in $C_(L|\sigma) \otimes C_*(L|\sigma)$. The second is carried, on a simplex λ of L , in $C_*(t\lambda) \otimes C_*(t\lambda)$. Consequently for $a = tS$ the homotopy*

$$\mathcal{D} = (t \otimes t)J_S + J_t S$$

has exactly the endpoints $(a \otimes a)\Delta_K$ and $\Delta_{K'} a$. If t approximates a member of an equicontinuous family of continuous maps h and both K, K' have mesh tending to zero, the two factors of every tensor term of $\mathcal{D}(\sigma)$ lie in a uniformly shrinking neighborhood of $h(|\sigma|)$.

Proof. For the first equation, the two chain maps have the displayed nested tensor carriers. The augmented chain complex of the tensor of the two subdivided simplices is acyclic: a contraction of a simplex induces a contraction of its tensor square. Both maps send each vertex to its vertex tensor square. The proof of Lemma 7.1, applied to these augmented chain subcomplexes, gives J_S . The same argument on the tensor square of the image simplex gives J_t , including simplicial maps which are not order preserving. Substitution proves the endpoint equation for $\mathcal{D}$.

For the last assertion, let ω be a common continuity modulus for the maps h , let the original source mesh be η , and the target mesh be η' . The simplicial approximation property puts each full target simplex occurring in t over a subdivided simplex inside the η' -neighborhood of the image of that subdivided simplex. All subdivided source simplices used above lie in the original σ . Thus both factors are within distance $\eta' + \omega(\eta)$ of $h(x)$, for any selected $x \in |\sigma|$. This tends to zero uniformly for the family. The target chain models and all existing endpoint maps are unchanged. $\square$

8. A FIXED CARRIER FOR A HYPOTHETICAL ACTION

Assume that $\mathbb{Z}_p$ acts effectively on a connected manifold M without boundary. The zero-dimensional case is immediate, so write $n = \dim M \geq 1$. Put $H_j = p^j \mathbb{Z}_p$.

Lemma 8.1. *The action has a point with trivial stabilizer.*

Proof. Every stabilizer is closed. A nonzero closed subgroup of $\mathbb{Z}_p$ is H_j for some j : an element of minimum p -adic valuation j has integer multiples dense in H_j , and every element of the subgroup lies in H_j . Regard $\mathbb{Z} \subset \mathbb{Z}_p$ as a discrete group. If every stabilizer were nonzero, its orbits would be finite. The finite-orbit theorem of Kaul and Yang, in the form [Rob75, Theorem 2], would give this restriction finite image. Some integer $m > 0$ would therefore act trivially on M . The kernel of the continuous $\mathbb{Z}_p$ -action is closed and would contain $\overline{m\mathbb{Z}} = H_{v_p(m)}$, contrary to effectivity. $\square$

Choose coordinates sending this point to the origin and whose image contains $\overline{B(0, 4)}$. Unless specified otherwise, balls in this section are Euclidean balls in this chart. Joint continuity on $\mathbb{Z}_p \times \overline{B(0, 2)}$ allows us to choose $H = H_j$ and a constant $0 < D < 1/64$ such that hx remains in the chart for $h \in H$ and $x \in \overline{B(0, 2)}$, and

$$(8.1) \quad |hx - x| \leq D \quad (h \in H, x \in \overline{B(0, 2)}).$$

The invariant open set $\Omega = HB(0, 2)$ lies in $B(0, 2 + D)$. Its H -displacements are at most $2D$: if $x = h_0 x_0$ with $x_0 \in B(0, 2)$, both x and hx are within D of x_0 .

Using normalized Haar measure, define the coordinate average

$$(8.2) \quad F_0(x) = \int_H hx \, d\mu(h), \quad x \in \Omega.$$

Uniform continuity on compact subsets proves continuity; Haar invariance proves $F_0(hx) = F_0(x)$. Moreover,

$$(8.3) \quad |F_0(x) - x| \leq 2D, \quad F_0(x) = x \quad \text{if } x \in \text{Fix}(H).$$

Choose $0 < \rho_+ < 1/4$ with $\overline{B(0, \rho_+)} \cap \text{Fix}(H) = \emptyset$, and put

$$(8.4) \quad Z^+ = F_0^{-1}(\overline{B(0, \rho_+)}).$$

This set is invariant and misses $\text{Fix}(H)$. It is compact because $|x| \leq \rho_+ + 2D$ on it, and this closed preimage lies inside a compact subset of $B(0, 2) \subset \Omega$.

Fix $0 < \rho < \rho_+$ and an oriented pinch $s_\rho : \mathbb{R}^n \rightarrow S^n$ which is a homeomorphism from $B(0, \rho)$ to $S^n \setminus \{\infty\}$ and is constant outside that ball. Define

$$(8.5) \quad f = s_\rho F_0 \text{ on } \Omega, \quad f = \infty \text{ off } \Omega.$$

The set on which f differs from ∞ lies in $B(0, \rho + 2D)$, so this extension is continuous. Extend f to the one-point compactification of coordinate space. This sphere map has degree one: the homotopy

$$(8.6) \quad s_\rho((1-v)F_0(x) + vx), \quad 0 \leq v \leq 1,$$

extended by ∞ , joins it to the chart pinch. By (8.3), this entire homotopy is constant outside one fixed compact subset of the chart. In particular, f is surjective and Z^+ is nonempty.

Put $H' = H_{j+1}$. Every proper closed subgroup of H is contained in H' , so $C_p = H/H'$ acts freely on Z^+/H' . The quotient is compact metrizable, and its free finite action is a covering action. Indeed, one can shrink a neighborhood of a point until its finitely many translates are disjoint. Choose a finite trivializing cover $\{U_a\}_{a=1}^m$ of Z^+/H and a subordinate partition of unity with supports inside the trivializing sets. In each lifted trivialization, multiply the C_p -valued sheet coordinate by the square root of the corresponding partition function, and extend by zero. These components give an equivariant map

$$(8.7) \quad \tilde{\rho} : Z^+ \longrightarrow T = S^{2m-1} \subset \mathbb{C}^m, \quad \tilde{\rho}(hx) = \chi(h)\tilde{\rho}(x),$$

where $\chi : H \rightarrow C_p \subset S^1$ is the quotient character. The squared absolute values of these components sum to one. Scalar multiplication makes T a fixed compact free isometric C_p -space.

Give the target sphere a round metric in which $y = s_\rho(0)$ and ∞ are antipodal. Fix

$$(8.8) \quad 0 < r < R < t < u < d(y, \infty), \quad Z_0 = \{z : d(y, z) \geq R\}, \quad Z = T \times Z_0.$$

The inverse image of $\overline{B(y, u)}$ under f is compactly contained in $\text{int } Z^+$. Thus every detector label needed below is defined on one fixed neighborhood of this inverse image. Only after fixing these spaces, maps, and buffers do we choose

$$(8.9) \quad K_i = H_{\ell_i} \subset H', \quad \ell_i \longrightarrow \infty, \quad G_i = H/K_i, \quad P_i = H'/K_i.$$

Since $K_i \subset \ker \chi$, we also write χ for the induced character of G_i . The K_i -displacements on $HB(0, 2)$ tend uniformly to zero. Neither T nor the trivializing cover used in the divisibility theorem varies with i .

9. COMMON CHAIN MODELS AND FINITE QUOTIENT RELATIONS

Put $W = S^n \times \mathbb{CP}^2$ and $N = n + 4$; all control maps ignore the second factor. Equip W with a product Riemannian metric. For the first coordinate retain the ordinary Euclidean chart on a neighborhood of $\overline{B(0, 3)}$; all points moved by the completed sphere maps, and all nonconstant tracks considered below, lie in that chart. Use additional sphere charts in the identity region near infinity, and a finite atlas on $\mathbb{CP}^2$. On the compact chart regions in use, choose a positive common buffer and chart and inverse-chart Lipschitz constants bounded by $A \geq 1$. A single increasing modulus ω , with $\omega(s) \rightarrow 0$ as $s \rightarrow 0$, may be used for the compact families of completed maps defined below, the original action on its fixed compact saturation, and the fixed control maps on their domains.

Let Y_i be fine combinatorial triangulations of W , of mesh $\eta_i \rightarrow 0$. They may be chosen by subdividing one triangulation compatible with the finite PL cutting flag used later. Set

$$D_i = C_*(Y_i; \mathbb{Q}), \quad z_i \in D_{i,N},$$

where z_i is the oriented fundamental cycle. The based modules have one oriented simplex per basis element, located at its barycenter. Control estimates use the entire closed simplices.

Lemma 9.1. *The complexes D_i admit strict symmetric cap equivalences $\phi_i : D_i^{N-*} \rightarrow D_i$, inverses b_i , and both inverse homotopies, all with geometric propagation at most $C_N \eta_i$. On these unchanged target complexes an equicontinuous family of self-maps of W admits chain approximations with uniform full-simplex graph error tending to zero. Their diagonal comparison homotopies may be chosen with the same uniform graph control.*

Proof. Use the dual-block orientation convention of lemma 7.4. If $z_i = \sum_\rho \epsilon_\rho \rho$, fix its zero-cell orientation by $D\rho = \epsilon_\rho [b_\rho]$. Thus its geometric augmentation is ϵ_ρ . The signed dual-block identification is $J_r(\sigma^*) = (-1)^{r(N+1)} D\sigma$, for the shifted dual convention used above. Apply corollary 7.6 with empty boundary. Primal and dual subdivision, their carrier inverses, and the cap comparison are contained in a fixed number of stars, whose diameters are $O_N(\eta_i)$. The ordered Alexander–Whitney cap map and its graded average have the required coface-star carriers and top augmentation. The local flip homotopy identifies them, so the average remains a Poincaré equivalence with the stated inverse data.

To construct approximations on the same target triangulation, choose a Lebesgue number for its open-star cover and subdivide the source sufficiently finely for the common continuity modulus. For each h , choose a simplicial approximation $t_h : L \rightarrow Y_i$, and put $a_h = t_h S$, where S is subdivision. Every target simplex in $a_h \sigma$ lies within

$$(9.1) \quad \mu_i = C_A \{\eta_i + \omega(C_A \eta_i)\} \longrightarrow 0$$

of h of some point of the original simplex σ . For the identity, choose the last-vertex map; the simplex-carried composite with subdivision is exactly the identity.

The diagonal comparisons can be made on these same complexes, as in lemma 7.7. Cone $(S \otimes S) \Delta_{Y_i} - \Delta_L S$ in the tensor of the subdivision of each source simplex, obtaining J_S . Cone $(t_h \otimes t_h) \Delta_L - \Delta_{Y_i} t_h$ in the tensor of each target simplex, obtaining J_{t_h} . The tensor contraction is $s \otimes 1 + (i\varepsilon) \otimes s$, where s contracts a simplex to its augmentation vertex. Then

$$\mathcal{D}_h = (t_h \otimes t_h) J_S + J_{t_h} S$$

satisfies the exact equation

$$d\mathcal{D}_h + \mathcal{D}_h d = (a_h \otimes a_h) \Delta - \Delta a_h.$$

Both factors of every tensor term are near the same $h(x)$, uniformly in the family. Averaging the flip gives the comparison for the strict symmetric diagonal as well. $\square$

Choose a fixed smooth cutoff $\theta : \mathbb{R}^n \rightarrow [0, 1]$, equal to 1 on $B(0, 1)$ and to 0 off $B(0, 3/2)$. Complete each $h \in H$ individually to a sphere map by

$$(9.2) \quad \widehat{h}(x) = x + \theta(x)(hx - x)$$

on the support of θ , using the identity elsewhere and at infinity. These completions form an equicontinuous family of degree-one maps, as is seen by scaling θ . Their group relations will be realized by the homotopies below, and

$$(9.3) \quad f\widehat{h} = f.$$

On $B(0, 1)$ this follows from invariance of f ; outside that ball both points miss the nonconstant locus of f .

Choose representatives $h_g \in H$ of G_i , with $h_1 = 1$, and write $h_g h_h = h_{gh} k_{g,h}$, $k_{g,h} \in K_i$. Define the product track

$$(9.4) \quad L_{g,h}(x, v) = (1 - v) \widehat{h}_g \widehat{h}_h(x) + v \widehat{h}_{gh}(x),$$

in Euclidean coordinates, with the constant track at ∞ . Let $d_i \rightarrow 0$ bound the K_i -displacements on the fixed compact chart saturation and put $\beta_i = \omega(Ad_i)$. If $|x| \leq 1/2$, all cutoff factors are

one, and this track is within β_i of $h_{gh}x$. If $|x| \geq 1/2$, every point of the segment is within $2D$ of x and has norm at least $1/2 - 2D > \rho + 2D$. Hence its f -value equals ∞ . These tracks are uniformly small in sphere control. On the buffered detector domain their graph control is of type gh .

Here a matrix has *graph type* g on that domain if every nonzero entry from x to x' has $f(x')$ approaching $f(x)$ and

$$d_T(\tilde{\rho}(x'), \chi(g)\tilde{\rho}(x)) \rightarrow 0$$

uniformly. Graph type describes support before sheet translation; the compensated maps will be morphisms over T .

Apply lemma 9.1 to $\hat{h}_g \times 1$ to obtain $a_{g,i} : D_i \rightarrow D_i$, with $a_{1,i} = 1$ exactly. Since these maps have degree one and there are no $(N+1)$ -chains,

$$(9.5) \quad a_{g,i}z_i = z_i.$$

All matrices below use this one target complex; auxiliary subdivisions used to fill cycles do not change it or its cap data.

Lemma 9.2 (Product homotopies with prescribed endpoints). *There are homotopies $B_{g,h}$ such that*

$$(9.6) \quad a_g a_h - a_{gh} = dB_{g,h} + B_{g,h}d.$$

Their f -propagations tend uniformly to zero, and on the buffered detector domain they have graph type gh , uniformly in i, g, h .

Proof. For $F = a_g a_h$ and $G = a_{gh}$ define, recursively on each oriented r -simplex,

$$\gamma_\sigma = F\sigma - G\sigma - B_{r-1}(\partial\sigma), \quad B_r\sigma = \text{Fill}(\gamma_\sigma).$$

The residual is a cycle by the identities on the faces; in degree zero it has augmentation zero. Use lemma 7.3 in the following carriers. The filling lemma gives the displayed homotopy equation. In top degree its local cycle assertion gives $B_N = 0$.

We record the uniform estimates, since the number of quotient elements and the auxiliary subdivision counts are unbounded. With μ_i as in (9.1), set

$$\delta_i = 2A\{\eta_i + \mu_i + \omega(A\mu_i) + \omega(A\eta_i) + \beta_i\} \rightarrow 0.$$

In the bounded sphere chart take the convex hull of the first-coordinate tracks $L_{g,h}(\sigma \times I)$. These hulls are nested under faces; a $C_A\delta_i$ enlargement contains the full endpoint chains. In the second factor use a coordinate-convex ball about the projection of σ . At each degree enlarge these product carriers to contain the already chosen face fillings. Chart changes enlarge a width w by at most A^2w , and the filling lemma adds one target mesh. For a sufficiently large fixed constant $c_{N,A}$, the recurrence

$$w_{-1} = 0, \quad w_r = A^2(w_{r-1} + c_{N,A}\delta_i) + 2A\eta_i = (c_{N,A}A^2\delta_i + 2A\eta_i) \sum_{j=0}^r A^{2j}$$

is sufficient. Thus $w_r \leq C_{N,A}\delta_i \rightarrow 0$ for $r \leq N$. The first-coordinate carriers remain in the Euclidean chart. For outer source cells their diameter may be $O(D)$, while the additional enlargement tends to zero. The estimate below places these carriers in $f^{-1}(\infty)$. The second-coordinate carriers and the inner first-coordinate carriers have diameters tending to zero and eventually fit within their fixed chart buffers. Filling may use different auxiliary cones at different cells and indices, but its exact base identity returns each fill to Y_i and preserves all previous endpoints and face fills.

Let x be the first coordinate of the barycenter of a source simplex. If $|x| \leq 3/8$, its first-coordinate projection eventually lies in $B(0, 1/2)$. The first-coordinate support of its filling lies within

$$e_i = \beta_i + \omega(A\eta_i) + \delta_i + w_N \longrightarrow 0$$

of $h_{gh}x$. On the detector domain, $\tilde{\rho}(h_{gh}x) = \chi(gh)\tilde{\rho}(x)$, giving the required graph type. For $|x| \geq 3/8$, the hull and its fills are within $2D + C_{N,A}\delta_i$ of x . The fixed margin

$$3/8 - \rho - 4D > 1/16$$

keeps their entire support outside the nonconstant locus of f . Consequently every such f -value is ∞ for all sufficiently large i . Near infinity all completed maps and tracks are the identity; small product-coordinate balls handle that region, with a switch in an identity overlap. Finally apply the fixed continuity moduli of the control maps to these carrier bounds. The constants depend on N and the fixed atlas, and not on i, g, h or the ranks. $\square$

The diagonal homotopy of [lemma 9.1](#), together with (9.5), gives

$$d(\mathcal{D}_g z_i) = (a_g \otimes a_g)\Delta z_i - \Delta z_i.$$

Right slant, reindexed as a degree-one map $U_g : D^{N-*} \rightarrow D$, and averaging yield a local homotopy with

$$(9.7) \quad a_g \phi a_g^* - \phi = dU_g + U_g d_{D^{N-*}}.$$

Both factors of each tensor term are near one target point, so U_g is local over W , uniformly in g . To obtain the adjoint relation put $a = a_g$ and $b = a_{g^{-1}}$. The identity

$$a\phi - \phi b^* = a\phi(1 - a^*b^*) + (a\phi a^* - \phi)b^*$$

uses only a bounded number of primitive maps. If $ba - 1 = dT + Td$, the shifted dual homotopy is $K_r = (-1)^{r+1}T_{N-r-1}^*$. Thus

$$(9.8) \quad V_g = -a\phi K + U_g b^*, \quad dV_g + V_g d_{D^{N-*}} = a_g \phi - \phi a_{g^{-1}}^*.$$

The product homotopy T has graph type 1, as does its transpose; U_g is local; and b^* has graph type g . Hence V_g has graph type g , uniformly on the buffered region. Let $\epsilon_i \rightarrow 0$ be a common bound for the f -errors, detector graph errors, cell oscillations, and duality data in these formulas.

10. COMPRESSION AND EXTERIOR CONTROL

Let $F_i \subset D_i$ be the subcomplex generated by simplices whose closed simplices miss $f^{-1}(\overline{B(y, t)})$. Define

$$(10.1) \quad C_i = D_i/F_i, \quad \pi_i : D_i \rightarrow C_i, \quad \sigma_i : C_i \rightarrow D_i, \quad Q_i = 1 - \sigma_i \pi_i,$$

where σ_i is the degreewise based section and Q_i is the degreewise projection onto F_i . We suppress i in matrix formulas. The quotient differential is $\hat{d} = \pi d\sigma$, and $\pi dQ = 0$.

If $\kappa_i \rightarrow 0$ bounds the oscillation of f on each closed cell, retained labels have distance at most $t + \kappa_i$ from y , whereas far labels have distance greater than t . Every retained cell and every cell reached by the bounded-length words used below eventually lies in the fixed detector domain. Label C_i by $(\tilde{\rho}, f)$.

Lemma 10.1 (Factorization through the exterior band). *Fix a maximum word length L . Consider products of at most L of the primitive maps and homotopies constructed above, their shifted duals, and the based projections and sections, with control errors bounded by ϵ_i . Every*

such product with retained source and target that contains Q or Q^* factors through a based module with labels in

$$(10.2) \quad t - L\epsilon_i \leq d(y, f(x)) \leq t + \kappa_i + L\epsilon_i.$$

For a scalar local word both factors are asymptotically small. For a graph word, the corresponding sheet translations make both factors small equivariant maps. In either case the intermediate object is supported in the fixed exterior Z .

Proof. Follow a nonzero summand through its intermediate basis elements. At the far projection its f -distance exceeds t , and every primitive factor changes that coordinate by at most ϵ_i . Both endpoints are retained. Restrict the far basis to the elements reached by the prefix of the word and contributing to its suffix. Cutting at this basis gives an exact matrix factorization, with unchanged coefficients, and proves (10.2). The same restriction to the contributing paths puts every intermediate label needed for the factors in the buffered detector domain. The band is eventually contained in (R, u) , so the cut object is supported in Z .

For scalar words each factor has propagation bounded by a sum of at most L primitive errors. For graph words include all free sheets and give sheet s over x the label $(s\tilde{\rho}(x), f(x))$. A graph- g map is accompanied by $s \mapsto sg^{-1}$; its target detector label then approaches $sg^{-1}\chi(g)\tilde{\rho}(x) = s\tilde{\rho}(x)$. At the cut divide the total translation into the prefix and suffix translations. Both factors are small and equivariant. The intermediate basis is a free G_i -set, invariant under strict left translation. Finite sums factor through direct sums of these witnesses; an unbounded number of summands at index i does not change the maximum propagation bound. $\square$

Put

$$(10.3) \quad A_g = \pi a_g \sigma, \quad \hat{\phi} = \pi \phi \pi^*, \quad \hat{b} = \sigma^* b \sigma.$$

For a continuous control map $c : W \rightarrow S^n$, let $\mathbf{m}_c$ denote the scalar class of (D_i, ϕ_i) with labels c and quadratic structure $\phi_i/2$ in $L_N(\mathcal{B}_{1,Z_0}(S^n))$.

Proposition 10.2. *The sequence $(C_i, \hat{d}_i, \hat{\phi}_i)$ is Poincaré in $\mathcal{B}_{1,Z}(T \times S^n)$. Its finite quotient product and adjoint relations have uniform graph carriers and controlled exterior factorizations after their sheet translations. Moreover $A_1 = 1$ exactly. After forgetting the T -coordinate, it represents the scalar manifold germ $\mathbf{m}_f$ in $L_N(\mathcal{B}_{1,Z_0}(S^n))$.*

Proof. We compute the defects explicitly and apply lemma 10.1 to each of them. For a compressed operator $\hat{\alpha} = \pi \alpha \sigma$, insertion of $\sigma \pi = 1 - Q$ gives

$$(10.4) \quad \hat{\alpha} \hat{\beta} - \hat{\alpha} \beta = -\pi \alpha Q \beta \sigma,$$

$$(10.5) \quad \hat{d} \hat{a} - \hat{a} \hat{d} = \pi a Q d \sigma,$$

$$(10.6) \quad A_g A_h - A_{gh} - \hat{d} \hat{B} - \hat{B} \hat{d} = -\pi a_g Q a_h \sigma + \pi B Q d \sigma, \quad \hat{B} = \pi B_{g,h} \sigma.$$

The last identity uses (9.6). Each error has the required graph type, and every translated error has a controlled factorization through the band object of lemma 10.1.

The map $\hat{\phi}$ is strictly symmetric and satisfies the chain-map equation before exterior quotienting, since π and its shifted dual are chain maps. For the inverse,

$$(10.7) \quad \hat{\phi} \hat{b} = \pi \phi b \sigma - \pi \phi Q^* b \sigma,$$

$$(10.8) \quad \hat{b} \hat{\phi} = \sigma^* b \phi \pi^* - \sigma^* b Q \phi \pi^*.$$

To check the chain and inverse defects, write

$$e = d\sigma - \sigma \hat{d} = Q d \sigma, \quad e^\vee = \hat{d} \sigma^* - \sigma^* \delta, \quad e_r^\vee = (-1)^{r+1} e_{N-r+1}^*,$$

where δ and $\widehat{\delta}$ are the shifted dual differentials. In particular $e^\vee$ factors through Q^* . Since b is a chain map,

$$\widehat{\delta}b - b\widehat{\delta} = e^\vee b\sigma + \sigma^* b e.$$

Let the local inverse homotopies be $\phi b - 1 = dH + Hd$ and $b\phi - 1 = \delta H' + H'\delta$. Direct compression gives

$$\begin{aligned}\widehat{\phi}b - 1 &= \widehat{d}(\pi H\sigma) + (\pi H\sigma)\widehat{d} + \pi H e - \pi\phi Q^* b\sigma, \\ \widehat{b}\widehat{\phi} - 1 &= \widehat{\delta}(\sigma^* H'\pi^*) + (\sigma^* H'\pi^*)\widehat{\delta} - e^\vee H'\pi^* - \sigma^* b Q\phi\pi^*.\end{aligned}$$

Every residual contains Q or Q^* and has bounded word length. The band factorization lemma makes these residuals zero in the exterior quotient. Hence $\widehat{b}$ is a chain homotopy inverse to $\widehat{\phi}$ there. Its quadratic representative is $\widehat{\phi}/2$, with higher components zero.

For the adjoint homotopy (9.8), use $\widehat{V}_g = \pi V_g \pi^*$. The chain-map equations for π and π^* give

$$\begin{aligned}A_g \widehat{\phi} - \widehat{\phi} A_{g-1}^* - \widehat{d}\widehat{V}_g - \widehat{V}_g \widehat{\delta} \\ = -\pi a_g Q\phi\pi^* + \pi\phi Q^* a_{g-1}^* \pi^*.\end{aligned}$$

Both residuals have graph type g . Their translated band factorizations prove the relation in the sheet-labeled category. All these formulas use one fixed maximum word length. Finally, $A_1 = \pi\sigma = 1$. If $\widehat{z} = \pi z_i$, the orientation check is the exact equation

$$(10.9) \quad A_g \widehat{z} - \widehat{z} = -\pi a_g Q z_i.$$

Its right-hand side is supported in the exterior band and has no component on retained top cells whose labels lie in $\overline{B(y, R)}$, for all sufficiently large indices.

After forgetting T , the whole complex D_i can be labeled by f . The far subcomplex F_i is Z_0 -supported. The projection Q factors through F_i , and the chain defect of σ is $Qd\sigma$. Consequently π and σ are inverse chain equivalences in $\mathcal{B}_{1, Z_0}(S^n)$. Transporting the symmetric cycle gives precisely $\pi\phi\pi^* = \widehat{\phi}$, and transporting its quadratic cycle gives the half of this map. Thus the scalar image is $\mathfrak{m}_f$. $\square$

11. THE GRAPH CORNER AND ITS SCALAR IMAGE

We now place the compressed graph maps in the category in which their displacements cancel. Give

$$\mathcal{C}_i = C_i \otimes \mathbb{Q}[G_i]$$

the free left basis action. For a retained basis simplex c , let x_c be the first coordinate of its barycenter. Label its copy on sheet s by

$$(11.1) \quad \ell(c \otimes s) = (\chi(s)\tilde{\rho}(x_c), f(x_c)).$$

The action on T is through $G_i/P_i = C_p$. Adding sheets defines a duality-preserving functor from the scalar category over $T \times S^n$: a local scalar matrix is copied on each sheet. Since the T -action is isometric, its propagation is unchanged. The functor also preserves exterior support.

Write $q_i = |G_i|$, and set

$$(11.2) \quad \Phi_i = \widehat{\phi}_i \otimes 1, \quad E_i = q_i^{-1} \sum_{g \in G_i} A_{g,i} \otimes R_{g-1}.$$

Here R_{g-1} sends sheet s to sg^{-1} . A graph- g entry has target T -label tending to $sg^{-1}\chi(g)\tilde{\rho}(x) = s\tilde{\rho}(x)$. Thus graph displacement and sheet translation cancel in the maps, the product and adjoint homotopies, and the exterior factorizations.

The compressed relations therefore give

$$(11.3) \quad E^2 \simeq E, \quad E\Phi \simeq \Phi E^*$$

in $K^b(\mathcal{B}_{\mathbf{G},Z}(T \times S^n))$. A representative of the product homotopy is

$$q_i^{-2} \sum_{g,h \in G_i} \hat{B}_{g,h} \otimes R_{(gh)^{-1}}.$$

Although the number of summands grows, propagation is bounded by the maximum of their uniform bounds. The residual factors through the direct sum of the band objects. The averaged adjoint homotopy is controlled in the same way. The uniform propagation bounds and fixed degree range define morphisms and homotopies in the sequence category.

Apply [lemma 5.1](#) to split E in $K^b(\text{Kar } \mathcal{B}_{\mathbf{G},Z}(T \times S^n))$ and equip its image with the normalized duality

$$(11.4) \quad D \xrightarrow{\iota} \mathcal{C} \xrightarrow{r} D, \quad \phi_D = q^{-1} r \Phi r^*.$$

The lemma gives its inverse and quadratic structure. Ultimate completion invariance returns the resulting class to the original quotient:

$$(11.5) \quad \mathfrak{d} \in L_{n+4}(\mathcal{B}_{\mathbf{G},Z}(T \times S^n)).$$

Let U forget equivariance and the T -coordinate, retaining sphere control. It maps the exterior to $Z_0 = \{z : d(y, z) \geq R\}$ and preserves duality. Every A_g and all homotopies used above are asymptotically small in the retained coordinate. Apply [lemma 5.2](#). Let j_1 and π_1 be inclusion and projection of the identity sheet, and write $q = (q_i)$. The comparison maps are

$$v = E j_1, \quad u = q \pi_1 E, \quad w = u \iota, \quad z = r v.$$

The maps w and z are mutually inverse chain equivalences and satisfy $w \phi_D w^* \simeq \hat{\phi}$. The quadratic classes agree because the averaged primitive of this symmetric homotopy, divided by two, identifies their quadratic cycles. Consequently

$$(11.6) \quad U_* \mathfrak{d} = \mathfrak{m}_f \quad \text{in } L_{n+4}(\mathcal{B}_{\mathbf{1},Z_0}(S^n)).$$

12. EVALUATION OF THE ITERATED BOUNDARY

Throughout this section $N = n + 4$ and $W = S^n \times \mathbb{CP}^2$, with its product orientation. All control maps ignore the second factor. We use the right-slant and shifted-dual conventions of [§5.1](#). In particular, an invariant tensor cycle t has quadratic representative $t/2$.

12.1. The fixed boundary homomorphism. Keep the point y , radii $r < R$, and exterior $Z = T \times \{z : d(y, z) \geq R\}$ fixed. Set

$$A = T \times (S^n \setminus B(y, r)), \quad B = T \times \overline{B(y, r)}, \quad Y = T \times S_r(y).$$

The exterior lies in A and is positively separated from B . The exterior Mayer–Vietoris boundary therefore takes values in $\mathcal{A}_{\mathbf{G},Y}(T \times S^n)$. Relabel supported objects onto Y in a fixed tubular neighborhood, and identify $S_r(y)$ with an oriented S^{n-1} . Cut the latter into two fixed closed hemispheres; repeat on each successive equator. After n boundaries the control space is $T \times S^0$. Its separated components split its category, and we project onto the positively oriented component. This defines

$$(12.1) \quad \Delta_T : L_N(\mathcal{B}_{\mathbf{G},Z}(T \times S^n)) \longrightarrow L_4(\mathcal{A}_{\mathbf{G}}(T)).$$

For $n = 1$ the first boundary already reaches S^0 . Each interface is oriented as the boundary of the designated A -side. All identifications, hemispheres, and the final component are chosen once, before the index varies.

Performing the same cuts without T or the group action gives

$$\Delta_1 : L_N(\mathcal{B}_{1,Z_0}(S^n)) \longrightarrow L_4(\mathcal{A}_1(\text{pt})), \quad Z_0 = \{z : d(y, z) \geq R\}.$$

The target is the point-tail category $\mathcal{V}_1$. If U forgets the group action and T but retains the sphere coordinate, naturality of every localization boundary and support relabeling gives

$$(12.2) \quad U_* \Delta_T = \Delta_1 U_*.$$

Tensoring with the positive quotient permutation representation likewise commutes with these boundaries.

12.2. Controlled Poincaré pairs. The following relative chain calculation supplies scalar cobordisms and geometric representatives of the localization boundaries.

Lemma 12.1 (Relative cap duality). *Let A be a compact oriented combinatorial m -manifold with boundary, triangulated with its boundary as a subcomplex. Write*

$$C = C_*(\partial A; \mathbb{Q}), \quad D = C_*(A; \mathbb{Q}), \quad j : C \longrightarrow D, \quad J = \text{Cone}(j).$$

Choose fundamental chains satisfying $dz_A = jz_{\partial A}$ and use the restriction of one ordered simplicial diagonal. The invariant relative tensors

$$t_D = \text{Sym } \Delta_A z_A, \quad t_C = \text{Sym } \Delta_{\partial A} z_{\partial A}, \quad \text{Sym} = (1 + T)/2,$$

satisfy $dt_D = j^{\otimes 2} t_C$ and $dt_C = 0$ exactly. Their right slants give the Poincaré-pair duality map

$$(12.3) \quad \Psi_r(\alpha, \beta) = \phi_{D,r} \alpha + (-1)^{r+1} j \phi_{C,r} \beta : J_r^{m-*} = D_{m-r}^* \oplus C_{m-r-1}^* \longrightarrow D_r.$$

For triangulations of mesh $\eta \rightarrow 0$, this map has a homotopy inverse and both inverse homotopies of geometric propagation $O_m(\eta)$. Consequently, labeling by any fixed continuous map on A gives asymptotically controlled Poincaré pairs, with the full relative quadratic structures $(t_D, t_C)/2$.

Proof. The exact relative tensor equation gives

$$d_D \phi_{D,r} - (-1)^r \phi_{D,r-1} d_D^* = j \phi_{C,r-1} j^*.$$

With $d_J(x, c) = (d_D x + j c, -d_C c)$, its shifted dual differential is

$$d_{J^{m-*},r} = (-1)^r \begin{pmatrix} d_D^* & 0 \\ j^* & -d_C^* \end{pmatrix}.$$

Substituting these two identities in (12.3) proves that Ψ is an exact chain map: the relative-boundary term cancels the j^* term.

Put $\mathcal{R} = D/C$ and let $q : J \rightarrow \mathcal{R}$ be the quotient map. Choose its degreewise based section $s : \mathcal{R} \rightarrow D$, and define $e : \mathcal{R}_r \rightarrow C_{r-1}$ by $j e = d_D s - s \bar{d}$, where $\bar{d}$ is the quotient differential. Then $d_C e + e \bar{d} = 0$, and

$$\tau = (s, -e) : \mathcal{R} \longrightarrow J, \quad q \tau = 1$$

is an exact chain map. Identify $\ker q$ with $\text{Cone}(1_C)$. Its contraction $k(x, c) = (0, x)$ gives

$$(12.4) \quad h = k(1 - \tau q), \quad dh + h d = 1 - \tau q.$$

The maps in this formula involve only identical simplices and their faces. Their shifted duals therefore have the same shrinking-star control.

The map $F = \Psi q^* : \mathcal{R}^{m-*} \rightarrow D$ is the relative cap map. Its image on a nonboundary cochain σ^* lies in the full closed coface star of σ . On a top simplex ρ , its value is $\epsilon_\rho/2$ times the sum of the first and last vertices, where ϵ_ρ is the coefficient of ρ in z_A . Its geometric augmentation is therefore ϵ_ρ . The relative tensor equation already proved makes F an exact chain map on

relative cochains. Lemma 7.5 consequently supplies a local inverse $b : D \rightarrow \mathcal{R}^{m-*}$ and local homotopies

$$Fb - 1 = dL + Ld, \quad bF - 1 = \delta_{\mathcal{R}}M + M\delta_{\mathcal{R}}.$$

The required relative dual-block carriers are those of Lemma 7.4.

To obtain an inverse for Ψ , set $B = q^*b$ and let

$$\kappa_r = (-1)^{r+1}h_{m-r-1}^*, \quad \delta_J\kappa + \kappa\delta_J = 1 - q^*\tau^*.$$

Then

$$\begin{aligned} \Psi B - 1 &= dL + Ld, \\ B\Psi - 1 &= \delta_J S + S\delta_J, \quad S = B\Psi\kappa + q^*M\tau^* - \kappa. \end{aligned}$$

Differentiating the last expression cancels the terms $q^*bF\tau^*$ and leaves $B\Psi - 1$. These are fixed finite compositions of maps supported in simplex stars. Their geometric propagation is $O_m(\eta)$, independently of rank, valence, and coefficient size. Applying a continuity modulus of the control map proves the assertion about control. Finally, §5.1 identifies the displayed relative invariant tensors and their halves with the full relative symmetric and quadratic structures, respectively. $\square$

Lemma 12.2 (Controlled unions). *Suppose a closed oriented combinatorial m -manifold is cut into two submanifolds W_A, W_B along their common boundary P . Use one triangulation and one ordered diagonal on all three spaces. Their relative fundamental tensors give a controlled Poincaré union whose full quadratic structure agrees with that of the closed manifold.*

Proof. Write the chains of W_A, W_B, P as A, B, P , with inclusions j_A, j_B , and choose

$$z_W = z_A + z_B, \quad dz_A = j_A z_P, \quad dz_B = -j_B z_P.$$

The corresponding invariant tensors satisfy $dt_A = j_A^{\otimes 2}t_P$, $dt_B = -j_B^{\otimes 2}t_P$, and $dt_P = 0$. Thus the two pairs of Lemma 12.1 have the same boundary complex and quadratic structure, with opposite signs.

Let $V = \text{Cone}((j_A, -j_B))$, and denote the inclusions of the A and B summands by e_A, e_B . Put $a = e_A j_A$, $b = e_B j_B$, and let $s_r : P_r \rightarrow V_{r+1}$ include the suspended interface summand. Then $ds + sd = a - b$. With the Koszul sign on tensor maps, set

$$\begin{aligned} K &= s \otimes a + b \otimes s, \quad K_{\text{sym}} = (K + T_V K T_P)/2, \\ t_V &= e_A^{\otimes 2}t_A + e_B^{\otimes 2}t_B - K_{\text{sym}}t_P. \end{aligned}$$

The second term of K has sign $(-1)^r$ when its first input factor has degree r . Hence $dK + Kd = a^{\otimes 2} - b^{\otimes 2}$ and $dt_V = 0$ exactly. This is the invariant-tensor form of the Poincaré union construction of [Ran80I, §3].

The collapse $p : V \rightarrow C_*(W; \mathbb{Q})$ adds the A and B summands and kills the suspended interface. Its kernel is $\text{Cone}(1_P)$, with contraction $k(x, c) = (0, x)$. Choose a based section r by selecting one copy of every shared simplex, and put $e = dr - rd$. The corrected section $\ell = r - ke$ satisfies

$$\begin{aligned} d\ell &= \ell d, \quad p\ell = 1, \\ d\{k(1 - \ell p)\} + \{k(1 - \ell p)\}d &= 1 - \ell p. \end{aligned}$$

Every summand of K contains s , so

$$(12.5) \quad (p \otimes p)t_V = \text{Sym } \Delta_W(z_A + z_B) = t_W.$$

In slant notation, $p\phi_V p^* = \phi_W$. If b_W is the local closed cap inverse, set $b_V = p^*b_W p$. The homotopies $\ell p \simeq 1$ give

$$\phi_V b_V \simeq \ell \phi_W b_W p \simeq \ell p \simeq 1, \quad b_V \phi_V \simeq p^* b_W \phi_W \ell^* \simeq p^* \ell^* \simeq 1.$$

All maps and homotopies use a fixed number of local compositions; they have propagation $O_m(\eta)$ before labeling, and propagation tending to zero after any fixed continuous labeling on W . Equation (12.5) identifies the full symmetric structures, and taking their halves identifies the full quadratic structures. The local mapping-cylinder cobordism then identifies their classes and all natural lower decorations. $\square$

12.3. Scalar homotopies and the standard pinch.

Lemma 12.3 (Homotopies of scalar control). *Let W be a compact closed oriented PL manifold of dimension m and $c : W \times [0, 1] \rightarrow X$ a fixed continuous map to a compact metric space. The two endpoint asymptotic manifold complexes are Poincaré cobordant, including their full rational quadratic structures. The equality persists after passage to any fixed exterior quotient of the control category.*

Proof. Use common endpoint triangulations of mesh $\eta_i \rightarrow 0$; local subdivision mapping cylinders allow this choice without changing their classes. Triangulate the prisms with time mesh $\lambda_i \rightarrow 0$, using one vertex order whose restrictions are the endpoint orders. Label every simplex at c of its barycenter. On the compact product, a simplex star has control-image diameter at most $\omega_c(C_m(\eta_i + \lambda_i)) \rightarrow 0$. The degree range is $[0, m + 1]$, independent of the time subdivision counts.

The cylinder fundamental chain $Z = (-1)^m z_W \times [0, 1]$ has boundary $j_1 z_W - j_0 z_W$ exactly. Its one ordered diagonal restricts to those of the two endpoints. Therefore its invariant tensor and the endpoint tensors satisfy the exact relative equation

$$dt_{W \times [0,1]} = j_1^{\otimes 2} t_1 - j_0^{\otimes 2} t_0.$$

Lemma 12.1, in dimension $m + 1$, supplies the controlled inverse and both inverse homotopies of this Poincaré pair. Taking halves gives the quadratic cobordism with exactly the prescribed endpoint structures. A fixed exterior quotient is a duality-preserving functor and sends this cobordism to a cobordism. The homotopy need not preserve the exterior: the quotient is applied to the completed cylinder, not separately to its time slices. The underlying ordinary symmetric construction is [Ran80II, Propositions 6.1–6.3]; the above prism carriers provide its control. $\square$

Recall the scalar class $\mathfrak{m}_f$ from §10. The fixed homotopy of f to the chart pinch s_ρ gives

$$(12.6) \quad \mathfrak{m}_f = \mathfrak{m}_{s_\rho} \quad \text{in } L_N(\mathcal{B}_{1,Z_0}(S^n))$$

by Lemma 12.3, applied after forgetting the auxiliary coordinate and equivariance.

Proposition 12.4 (Value of the standard pinch). *With the fixed boundary orientations,*

$$(12.7) \quad \Delta_1 \mathfrak{m}_{s_\rho} = [C_*(\mathbb{CP}^2; \mathbb{Q}), \phi_{\mathbb{CP}^2}] \quad \text{in } L_4(\mathcal{A}_1(\text{pt})).$$

The right-hand side denotes the constant sequence of Poincaré complexes of $\mathbb{CP}^2$ with its complex orientation. In particular, its tail signature is the constant sequence 1.

Proof. Keep all categorical cuts in the target fixed. Choose an oriented standard pinch $g : S^n \rightarrow S^n$ which identifies a source PL disk D and a collar with the first target ball and collar, carries a fixed PL hemisphere flag of ∂D to the target flag, and collapses the outer end of the complementary annulus to infinity. A point in the interior target ball has one positively oriented preimage, so g has degree one. Sphere maps of degree one are homotopic: for $n \geq 2$ use $\pi_n(S^n) \cong H_n(S^n)$, and for $n = 1$ use winding number [Hat02, Theorem 4.32 and §1.1]. Lemma 12.3 therefore replaces s_ρ by g , without changing any target cut or boundary map.

Triangulate W once respecting

$$W_B = D \times \mathbb{CP}^2, \quad W_A = (S^n \setminus \text{int } D) \times \mathbb{CP}^2,$$

their interface $\Sigma = \partial D \times \mathbb{CP}^2$, and every member of the finite subsequent hemisphere flag. Use common barycentric refinements and one ordered diagonal throughout. Lemmas 12.1 and 12.2 give compatible relative quadratic Poincaré pairs with union equal to the scalar manifold class. After labeling by g , the two pair targets are supported on $S^n \setminus B(y, r)$ and $\overline{B}(y, r)$, respectively, and their common boundary on $S_r(y)$. The propagation of every displayed map is bounded by $\omega_g(C_N \eta_i) \rightarrow 0$. Since $r < R$, the closed-ball and interface support categories embed fully faithfully in the exterior quotient by lemma 2.4. The representative formula for the localization boundary now gives the class of Σ , oriented as ∂W_A .

Apply the same relative construction to the two hemispheres of every successive interface. The common triangulation and diagonal make their boundary chains, orientations, and full quadratic structures compatible. The successive classes are

$$[S^{n-2} \times \mathbb{CP}^2], \dots, [S^0 \times \mathbb{CP}^2]$$

when $n \geq 2$; for $n = 1$ the first interface is already the last. The prescribed positive component is one positively oriented copy of $\mathbb{CP}^2$. This proves (12.7) before passing to lower decorations, and therefore in the stated ultimate group as well.

The middle rational homology of $\mathbb{CP}^2$ is generated by a projective line. Two transverse projective lines have one positive intersection with the complex orientation, so its form is $\langle 1 \rangle$. The half-unit quadratic representative has this same symmetrized form. Its signature is consequently 1 at every index. Combined with (12.6), this gives the integral tail equality

$$(12.8) \quad \sigma_{\text{tail}}(\Delta_1 \mathbf{m}_f) = [(1, 1, \dots)].$$

□

13. PROOF OF THE MAIN THEOREM

The construction gives the following realization.

Proposition 13.1 (Signature-one realization). *If $\mathbb{Z}_p$ acts continuously and effectively on a connected positive-dimensional manifold without boundary, there are a fixed compact free C_p -space T , finite cyclic groups G_i with index- p subgroups P_i , and a class $\alpha \in L_4(\mathcal{A}_{\mathbf{G}}(T))$ such that*

$$(13.1) \quad \sigma_{\text{tail}}(\alpha) = [(1, 1, \dots)].$$

Proof. The construction in §8 supplies T , the groups $G_i = H/K_i$, their subgroups $P_i = H'/K_i$, and the control map f . The common chain models and lemmas 9.2 and 10.1 and proposition 10.2 give Poincaré data, translated product and adjoint homotopies, and factorizations through the single exterior Z . Thus (11.3) holds in $\mathcal{B}_{\mathbf{G}, Z}(T \times S^n)$. Applying lemma 5.1 there gives the class $\mathfrak{d}$ of (11.5).

After forgetting equivariance and T , lemma 5.2 gives the scalar homotopy isometry (11.6). Set $\alpha = \Delta_T \mathfrak{d}$, using (12.1). Naturality (12.2) gives

$$U_* \alpha = \Delta_1 \mathbf{m}_f.$$

The scalar homotopy (12.6) and proposition 12.4 identify this class with the point-tail class of one positively oriented $\mathbb{CP}^2$. Its signature is 1. By (3.2), the signature of each extracted equivariant form equals the signature of its scalar image. Equation (12.8) therefore gives (13.1). □

Proof of theorem 1.1. A connected zero-dimensional manifold is a point, so suppose its dimension is positive. If it has boundary, its interior is dense and connected. Density follows from half-space charts; the interior of each boundary half-ball is connected, so each boundary point

adjoins only one interior component. Distinct components would then induce a separation of the whole manifold. Homeomorphisms preserve the interior and boundary, distinguished by degree- n local homology with $\mathbb{Z}/2$ coefficients. A homeomorphism fixing the dense interior fixes the whole manifold. Restriction to the interior therefore preserves effectivity, and it suffices to treat a manifold without boundary.

Suppose, for a contradiction, that the action is effective. [Proposition 13.1](#) gives $\alpha \in L_4(\mathcal{A}_G(T))$ satisfying (13.1). Since T is a fixed free C_p -space, [theorem 4.3](#) gives

$$(13.2) \quad \sigma_{\text{tail}}(\alpha) \in \frac{\prod_i p\mathbb{Z}}{\bigoplus_i p\mathbb{Z}} \subset \frac{\prod_i \mathbb{Z}}{\bigoplus_i \mathbb{Z}}.$$

Use [proposition 3.2](#) to choose one tail of finite equivariant forms representing the image of α after forgetting labels. Equation (13.1) makes their ordinary signatures 1 at every sufficiently large index, while (13.2) makes those same signatures divisible by p at every sufficiently large index. A common tail gives $1 \in p\mathbb{Z}$, a contradiction. $\square$

Proof of corollary 1.2. In dimension zero, effectivity forces G to be trivial. In positive dimension the preceding interior argument preserves effectivity, so assume the manifold has no boundary. Every compact subgroup acts effectively. By the compact p -adic reduction [[Lee97](#), [Pa19](#)], a non-Lie compact subgroup admitting an effective action would give an effective action of $\mathbb{Z}_p$ for some prime p . [Theorem 1.1](#) therefore implies that every compact subgroup of G is a Lie group.

Choose a compatible metric on the manifold, a nonempty relatively compact open chart V_M , and its compact closure K . The uniform Newman theorem gives $\varepsilon > 0$ on V_M , independent of the compact Lie group [[Pa19](#), Theorem (Newman)]. Joint continuity and compactness of K give an identity neighborhood V_G such that $d(x, gx) < \varepsilon$ for $x \in K$ and $g \in V_G$. Choose an open identity neighborhood W_G with compact closure contained in V_G . If a subgroup S lies in W_G , its closure is a compact subgroup contained in $\overline{W}_G \subset V_G$. It is Lie, and Newman forces its action to be trivial on the connected manifold. Effectivity then gives $S = \{1\}$. Thus G has no small subgroups. The Gleason–Yamabe theorem [[Gle52](#), [Yam53](#)], in the form stated in [[Gol10](#), Section 8], gives a finite-dimensional Lie structure with the original topology. Since G is second countable, it is a Lie group under the convention of §1. $\square$

DISCLAIMER

Various large language models (LLMs), including commercial models from Anthropic and OpenAI and open-source models, were used under human guidance at every stage of this project and for nearly every task, from understanding, planning and ideation to research, preparation and writing of the final draft. A custom LLM orchestration framework was implemented and used for the research phase. The core idea and Lean4 formalization were obtained by Claude Opus 5.5 agents. This private personal project had a constrained budget, total costs were below US\$1,000.

REFERENCES

- [BS01] P. Balmer and M. Schlichting, *Idempotent completion of triangulated categories*, J. Algebra **236** (2001), no. 2, 819–834. [doi:10.1006/jabr.2000.8529](#).
- [BM46] S. Bochner and D. Montgomery, *Locally compact groups of differentiable transformations*, Ann. of Math. (2) **47** (1946), no. 4, 639–653. [doi:10.2307/1969226](#).
- [CP95] G. Carlsson and E. K. Pedersen, *Controlled algebra and the Novikov conjectures for K- and L-theory*, Topology **34** (1995), no. 3, 731–758. [doi:10.1016/0040-9383\(94\)00033-H](#).
- [Gle52] A. M. Gleason, *Groups without small subgroups*, Ann. of Math. (2) **56** (1952), no. 2, 193–212. [doi:10.2307/1969795](#).

- [Gol10] I. Goldbring, *Hilbert’s fifth problem for local groups*, Ann. of Math. (2) **172** (2010), no. 2, 1269–1314. doi:10.4007/annals.2010.172.1269.
- [Hat02] A. Hatcher, *Algebraic Topology*, Cambridge University Press, Cambridge, 2002.
- [Hil02] D. Hilbert, *Mathematical problems*, Bull. Amer. Math. Soc. **8** (1902), no. 10, 437–479. doi:10.1090/S0002-9904-1902-00923-3.
- [Lee97] J. S. Lee, *Totally disconnected groups, p -adic groups and the Hilbert–Smith conjecture*, Commun. Korean Math. Soc. **12** (1997), no. 3, 691–699.
- [Mal97] I. Maleshich, *The Hilbert–Smith conjecture for Hölder actions*, Russian Math. Surveys **52** (1997), no. 2, 407–408; Russian original, Uspekhi Mat. Nauk **52** (1997), no. 2(314), 173–174. doi:10.1070/RM1997v052n02ABEH001792.
- [Mar99] G. J. Martin, *The Hilbert–Smith conjecture for quasiconformal actions*, Electron. Res. Announc. Amer. Math. Soc. **5** (1999), 66–70. doi:10.1090/S1079-6762-99-00062-1.
- [Mj12] M. Mj, *Pattern rigidity and the Hilbert–Smith conjecture*, Geom. Topol. **16** (2012), no. 2, 1205–1246. doi:10.2140/gt.2012.16.1205.
- [MZ55] D. Montgomery and L. Zippin, *Topological Transformation Groups*, Interscience Publishers, New York–London, 1955.
- [New31] M. H. A. Newman, *A theorem on periodic transformations of spaces*, Quart. J. Math. Oxford Ser. **2** (1931), 1–8. doi:10.1093/qmath/os-2.1.1-a.
- [Pa13] J. Pardon, *The Hilbert–Smith conjecture for three-manifolds*, J. Amer. Math. Soc. **26** (2013), no. 3, 879–899. doi:10.1090/S0894-0347-2013-00766-3.
- [Pa19] J. Pardon, *Totally disconnected groups (not) acting on two-manifolds*, in *Breadth in Contemporary Topology*, D. T. Gay and W. Wu (eds.), Proc. Sympos. Pure Math. **102**, American Mathematical Society, Providence, RI, 2019, 187–193. doi:10.1090/pspum/102/01809.
- [Ran80I] A. A. Ranicki, *The algebraic theory of surgery. I. Foundations*, Proc. London Math. Soc. (3) **40** (1980), no. 1, 87–192. doi:10.1112/plms/s3-40.1.87.
- [Ran80II] A. A. Ranicki, *The algebraic theory of surgery. II. Applications to topology*, Proc. London Math. Soc. (3) **40** (1980), no. 2, 193–283. doi:10.1112/plms/s3-40.2.193.
- [Ran89] A. A. Ranicki, *Additive L -theory, K -Theory* **3** (1989), no. 2, 163–195. doi:10.1007/BF00533377.
- [Ran92] A. A. Ranicki, *Lower K - and L -Theory*, London Math. Soc. Lecture Note Ser. **178**, Cambridge University Press, Cambridge, 1992.
- [RS97] D. Repovš and E. V. Ščepin, *A proof of the Hilbert–Smith conjecture for actions by Lipschitz maps*, Math. Ann. **308** (1997), no. 2, 361–364. doi:10.1007/s002080050080.
- [Rob75] J. W. Roberts, *Pointwise finite families of mappings*, Canad. Math. Bull. **18** (1975), no. 5, 767–768. doi:10.4153/CMB-1975-134-x.
- [Ros06] D. Rosenthal, *Continuous control and the algebraic L -theory assembly map*, Forum Math. **18** (2006), no. 2, 193–209. doi:10.1515/FORUM.2006.012.
- [She24] E. Shelukhin, *A symplectic Hilbert–Smith conjecture*, with an appendix by L. Polterovich, arXiv:2403.07195, 2024. https://arxiv.org/abs/2403.07195.
- [Smi41] P. A. Smith, *Transformations of finite period. III. Newman’s theorem*, Ann. of Math. (2) **42** (1941), no. 2, 446–458. doi:10.2307/1968910.
- [Wei13] C. A. Weibel, *The K -Book: An Introduction to Algebraic K -Theory*, Grad. Stud. Math. **145**, American Mathematical Society, Providence, RI, 2013.
- [Yam53] H. Yamabe, *A generalization of a theorem of Gleason*, Ann. of Math. (2) **58** (1953), no. 2, 351–365. doi:10.2307/1969792.
- [Yang60] C.-T. Yang, *p -adic transformation groups*, Michigan Math. J. **7** (1960), no. 3, 201–218. doi:10.1307/mmj/1028998428.

Email address: alex@dabrow.cc